



## Mini-review

## Advances in catalytic dehydrogenation and dehydrogenative coupling of ethanol to acetaldehyde, ethyl acetate and n-butanol

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## ARTICLE INFO

## Keywords:

Ethanol  
Dehydrogenation  
Dehydrogenative coupling  
Acetaldehyde  
Heterogeneous catalysis

## ABSTRACT

As a renewable biomass feedstock, bioethanol plays a crucial role in the chemical industry, particularly in the production of chemicals and high-value fuels. Ethanol can be selectively converted into acetaldehyde, ethyl acetate, and n-butanol by heterogeneous catalysis, with hydrogen as the sole byproduct. Hydrogen is also widely recognized as a promising clean energy carrier due to its high energy conversion efficiency and environmentally benign combustion product (water). This article provides a mini-review of the recent development of the catalytic dehydrogenation and dehydrogenative coupling of ethanol to acetaldehyde, ethyl acetate, and n-butanol over heterogeneous catalysts. The details include the typical catalysts, the catalyst design strategy and reaction mechanisms. The application potential of these catalysts is also explored. Finally, the main challenges facing the current technology and opportunities in replacing the fossil route were proposed.

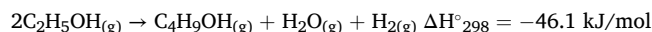
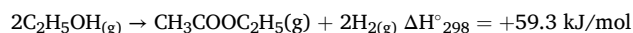
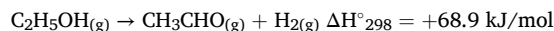
## 1. Introduction

In recent years, the excessive consumption of fossil fuels has led to an increasingly severe energy crisis, prompting countries worldwide to attach great importance to and actively promote biomass energy strategies. Unlike fossil fuels, the raw materials for ethanol production come from a wide range of sources, including various renewable biomass resources such as corn, sugarcane, straw, and forestry waste [1,2]. As the only carbon-containing renewable resource available today, biomass demonstrates significant potential in producing carbon-neutral fuels and key chemicals, and is gradually becoming an important pathway to replace corresponding fossil-based products [3–5].

Ethanol dehydrogenation and subsequent coupling can generate a range of high-value-added chemicals such as acetaldehyde, ethyl acetate, and n-butanol, which not only significantly enhance the economic value of ethanol, but also provide an important solution for efficient resource utilization and reducing environmental pollution [6–10]. As shown in Scheme 1, the typical reaction network of ethanol includes dehydrogenation to form acetaldehyde, which can subsequently undergo further coupling to produce ethyl acetate and n-butanol. (It should be noted that the Lebedev process for converting ethanol to butadiene is also an important route. However, as it has been extensively covered in

dedicated reviews [11–13], it will not be discussed in this review.) However, by acid catalysis or under high temperatures, ethanol may undergo side reactions, such as dehydration to form ethylene, diethyl ether and 1,2-diethoxyethane [14–16]. Additionally, at excessively high reaction temperatures (> 300 °C) and using highly active catalysts in breaking C–C bonds (nickel-based catalysts), ethanol fully decomposes to produce CO and CH<sub>4</sub> [17]. Due to the metastability of ethanol and acetaldehyde under most reaction conditions, controlling the reaction pathway and product selectivity remains a challenge.

The dehydrogenation of ethanol produces acetaldehyde, which may be accompanied by the production of by-products such as ethyl acetate and n-butanol. However, by designing the catalyst's composition and microstructure (to regulate its active sites) and combining this with optimized reaction conditions, ethyl acetate or n-butanol can be made the primary product, while acetaldehyde serves as a reaction intermediate [18–22]. Looking at the chemical reactions to form acetaldehyde, ethyl acetate and n-butanol from ethanol:



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Received 9 July 2025; Received in revised form 18 October 2025; Accepted 16 November 2025

Available online 19 November 2025

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The first two are endothermic reactions, limited by thermodynamic equilibrium, and require high temperatures and/or low pressures to increase conversion. In contrast, the synthesis of n-butanol is exothermic, possesses a significant thermodynamic driving force, and is virtually unrestricted by equilibrium constraints. However, the practical challenges lie in kinetic and selectivity control, necessitating multifunctional catalysts to manage the complex multi-step reaction pathway and prevent the formation of by-products.

Typically, the dehydrogenation of ethanol to acetaldehyde requires metal sites ( $\text{Cu}^0$ ) and appropriate Lewis acidic sites (such as  $\text{Cu}^+$ ,  $\text{Pt}^+$ ,  $\text{Ni}^+$ , etc.) to activate the O–H bond and the  $\alpha$ -C–H bond of ethanol. Simultaneously, neutral and inert supports (e.g.,  $\text{SiO}_2$ , dealuminated Beta zeolite, mesoporous carbon, etc.) are often employed to anchor and disperse these active centers, while leveraging their inert surfaces to suppress secondary reactions of the acetaldehyde product (such as aldol condensation). Typical reaction conditions involve atmospheric pressure and a temperature range of 200–300 °C [14,23–29]. The conversion of ethanol to ethyl acetate not only requires metal sites and Lewis acid sites to generate the acetaldehyde intermediate, but also additional Lewis acid sites (e.g.,  $\text{Cu}^+$ ,  $\text{CuCr}_2\text{O}_4$ ,  $\text{ZrO}_2$ ) to facilitate the coupling of acetaldehyde with ethanol and the subsequent desorption of the ethyl acetate product. This reaction is typically carried out at temperatures between 250–300 °C and a pressure greater than 0.5 MPa [30–32]. The conversion of ethanol to n-butanol relies on the synergistic interplay of multiple active sites within the catalyst. These include metallic sites capable of dehydrogenation and hydrogenation (e.g., Pt, Pd), Lewis acid sites that activate the carbonyl group and stabilize intermediates (e.g.,  $\text{Ca}^{2+}$  in hydroxyapatite), and basic sites that abstract the  $\alpha$ -hydrogen (e.g.,  $\text{PO}_4^{3-}$  in hydroxyapatite). The reaction is typically conducted at 250–350 °C under 2–3 MPa [33–35].

Clearly, there is an intersection between the catalytic active species and reaction conditions for the three products, suggesting potential competition in the reaction. Therefore, developing catalysts with high conversion rates and selectivity to precisely control the balance between the main and by-products remains a significant challenge.

In recent years, significant advancements have been made in the catalytic conversion of biomass ethanol into acetaldehyde, ethyl acetate, and n-butanol. In the review articles of this field, Sun et al. [6] and Nada et al. [36] briefly mentioned the reaction mechanism of the ethanol dehydrogenation reaction. Huang et al. [37] published a mini-review that systematically summarized the application of heterogeneous catalysts in ethanol dehydrogenation to acetaldehyde. Phung [38] focused

on the application of Copper-based (Cu-based) catalysts in the ethanol dehydrogenation reaction, examining the effects of copper valence state, copper particle size and dispersion, support, promoters, and reaction conditions. Pang et al. [39] highlighted representative catalysts for both oxidative and non-oxidative dehydrogenation of ethanol and analyzed the reaction mechanisms of typical catalysts. However, a comprehensive summary of the active species, activity-structure relationships, and reaction mechanisms of these catalytic systems for ethanol dehydrogenation and dehydrogenation coupling is still lacking. In this review, we discuss in detail over the past 15 years the research progress on the catalytic conversion of acetaldehyde, ethyl acetate, and n-butanol by ethanol dehydrogenation and dehydrogenation coupling reaction, including the structure-activity relationships, selectivity modulation, reaction mechanism, and application prospect of the catalysts. Finally, the review also summarizes the major challenges in the current studies, hoping to provide a reference for the subsequent research in this field.

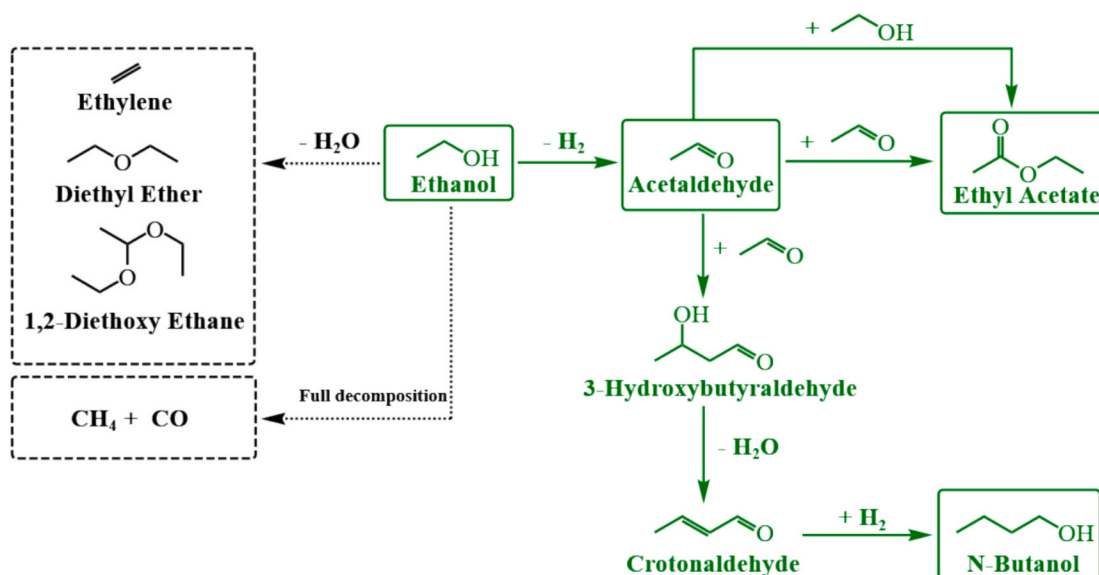
## 2. Non-oxidative dehydrogenation of ethanol to acetaldehyde

The traditional industrial methods for producing acetaldehyde mainly include ethylene oxidation (i.e., the Wacker-Hoechst process) [40–42] and oxidative dehydrogenation of ethanol [43]. In contrast, non-oxidative dehydrogenation of ethanol to acetaldehyde has gradually gained attention due to its environmental friendliness and economic performance [44]. In this process, ethanol passes over metal-supported catalysts (e.g., copper, gold, and silver) at 200–300 °C to selectively break the O–H and the  $\alpha$ -C–H bond in the ethanol molecule to efficiently produce acetaldehyde, while generating hydrogen as a clean byproduct (laboratory conditions). This review systematically analyzes the catalytic performance of various catalysts for ethanol dehydrogenation to acetaldehyde, as summarized in Table 1 and Table 2.

### 2.1. Active sites and reaction mechanism of Cu-based catalysts

Cu-based catalysts have shown significant potential in the ethanol dehydrogenation reaction due to the high activity of Cu species in O–H and C–H bonds oxidative cleavage and relatively limited activity in C–C bond cleavage reactions [45–50]. However, the key active sites on Cu-based catalysts for ethanol dehydrogenation are still debated, primarily due to their dynamic nature in catalytic reactions.

He et al. [23] demonstrated that via  $\text{N}_2\text{O}$ -induced in situ surface restructuring of Cu nanoparticles, the remarkable enhancement in



Scheme 1. Typical reaction network of ethanol.

**Table 1**Cu-based catalysts and metal oxide catalysts for dehydrogenation of ethanol to acetaldehyde.<sup>a</sup>

| Catalyst                               | Reaction conditions                                                                                              | Conv.  | Sel.   | Features                                                                                                                                      | Ref. |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------|------|
| Cu/HT600(10 wt % Cu)                   | N <sub>2</sub> , WHSV = 1.06 h <sup>-1</sup> , 300 °C, 1 atm, reaction 6 h                                       | 85.5 % | 97.7 % | CuAl <sub>2</sub> O <sub>4</sub> spinel structure inhibits copper agglomeration                                                               | [57] |
| Cu/KIT-6 (62 wt% Cu)                   | N <sub>2</sub> , WHSV = 0.87 h <sup>-1</sup> , 250 °C, atmospheric pressure                                      | 96.8 % | 99.8 % | Three-dimensional pore structure, high active surface area (920 m <sup>2</sup> /g), and metal dispersion (62.3 %), with abundant acidic sites | [45] |
| Cu@S-1 (2 wt% Cu)                      | N <sub>2</sub> , WHSV = 0.53 h <sup>-1</sup> , 250 °C, atmospheric pressure, TOS 110 h                           | 97.8 % | 91.8 % | copper nanoparticles (~1.8 nm) within S-1 zeolite pores, Maintains high Cu <sup>+</sup> species ratio (~70 %)                                 | [14] |
| Cu-MFI-AE (19 wt% Cu)                  | N <sub>2</sub> , WHSV = 0.53 h <sup>-1</sup> , 250 °C, atmospheric pressure                                      | 97.9 % | 93.1 % | Hierarchical urchin-like structure, Strong Cu – O – Si interface interaction                                                                  | [29] |
| (Cu/MFI)@MFI (5.6 wt% Cu)              | N <sub>2</sub> , WHSV = 0.53 h <sup>-1</sup> , 250 °C, atmospheric pressure, TOS 200 h                           | 95.7 % | 97.2 % | Maintains high activity and selectivity at 200 h                                                                                              | [28] |
| Cu@mSiO <sub>2</sub> (6.9 wt% Cu)      | N <sub>2</sub> , WHSV = 1.5 h <sup>-1</sup> , 240 °C, atmospheric pressure, reaction 1 h                         | 80.4 % | 93.1 % | Mesoporous Silica-Encapsulated Cu Nanoparticles; the Cu <sup>+</sup> /(Cu <sup>0</sup> + Cu <sup>+</sup> ) ratio                              | [27] |
| Cu-MFI-deCu (5 wt% Cu)                 | N <sub>2</sub> , WHSV = 0.64 h <sup>-1</sup> , 250 °C, atmospheric pressure, TOS 120 h                           | 87 %   | 95 %   | Highly dispersed Cu <sup>δ+</sup> (1 < δ < 2) species on the MFI support                                                                      | [47] |
| Cu/SiO <sub>2</sub> -HSG (2.13 wt% Cu) | N <sub>2</sub> , WHSV = 4.73 h <sup>-1</sup> , 255 °C, atmospheric pressure, reaction 1 h                        | 65 %   | >95 %  | ~2.5 wt% Cu, sub-nanometer clusters                                                                                                           | [63] |
| Cu@MFI-R (2 wt% Cu)                    | N <sub>2</sub> , WHSV = 0.55 h <sup>-1</sup> , 250 °C, atmospheric pressure, reaction 1 h                        | >96 %  | 98 %   | Add ethylenediamine and propylenediamine additives, Cu <sup>+</sup> species remain stable                                                     | [50] |
| Cu/ZnO@SiO <sub>2</sub> (30 wt% Cu)    | N <sub>2</sub> , WHSV = 3.2 h <sup>-1</sup> , 310 °C, 0.1 MPa, TOS 50 h                                          | 84.2 % | 99.1 % | Most Cu particles were dotted and isolated with fine ZnO                                                                                      | [24] |
| Cu-A5CZ-180                            | He, 320 °C, atmospheric pressure                                                                                 | 68.6 % | 94.3 % | Sol-gel method, Mixed oxide support containing various alumina ratios                                                                         | [64] |
| Cu/MC (10 wt% Cu)                      | N <sub>2</sub> , WHSV = 2.4 h <sup>-1</sup> , 280 °C, atmospheric pressure                                       | 72.7 % | 94 %   | Mesoporous carbon support, Unique pore structure enhances ethanol molecule enrichment capacity                                                | [49] |
| ZnO/Rb-MOR (3.2 wt% ZnO)               | N <sub>2</sub> , HEL FlowCAT flow reactor, ethanol = 0.330 mmol min <sup>-1</sup> , 400 °C, atmospheric pressure | 40 %   | 90 %   | ZnO achieves extremely low ethylene selectivity and long reaction lifetime (120 h).                                                           | [60] |
| V-Zr-La/SBA-15-HT                      | N <sub>2</sub> , WHSV = 22.9 h <sup>-1</sup> , 300 °C, atmospheric pressure                                      | 98 %   | 40.8 % | The monomeric V <sup>4+</sup> species (V=O) is the key active site for ethanol dehydrogenation.                                               | [61] |

<sup>a</sup> Reactions were performed in a fixed bed reactor if not specified: Conv. and Sel. are abbreviations for ethanol conversion and acetaldehyde selectivity, respectively.**Table 2**Single-atom catalysts for dehydrogenation of ethanol to acetaldehyde.<sup>a</sup>

| Catalyst                            | Reaction conditions                                                                            | Conv.  | Sel.   | Feature                                                                                                                                                                     | Ref. |
|-------------------------------------|------------------------------------------------------------------------------------------------|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| NiAu SAAs                           | Reaction gas composition: 2 % ethanol + He (carrier gas), 250 °C, atmospheric pressure         | 27 %   | 100 %  | Ni atoms are dispersed as single atoms on the Au substrate; Ni sites promote C–H and O–H bond dissociation, while Au sites provide selectivity and resistance to sintering. | [25] |
| Ni-Cu/SiO <sub>2</sub> (10 wt % Cu) | Reaction gas composition: 4 % ethanol + He (carrier gas), WHSV = 2.37 h <sup>-1</sup> , 300 °C | 81.9 % | 85.6 % | Addition of Ni significantly increased conversion but decreased acetaldehyde selectivity, with an increase in methane production.                                           | [26] |
| Cd-Cu/SiO <sub>2</sub> (10 wt % Cu) |                                                                                                | 77 %   | 94.9 % | Cd improved the stability and activity of the catalyst.                                                                                                                     |      |
| Pt-Cu/SiO <sub>2</sub> (10 wt % Cu) |                                                                                                | 66.7 % | 84.4 % | The introduction of Pt favors the formation of by-products such as diethyl ether.                                                                                           |      |

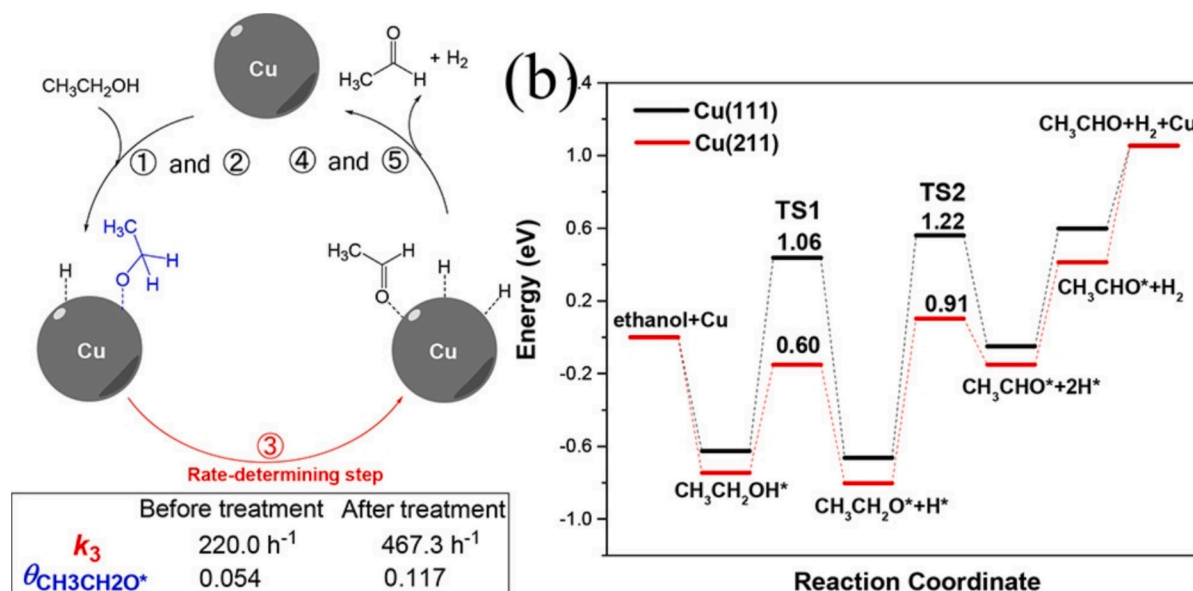
<sup>a</sup> Reactions were performed in a fixed bed reactor if not specified: Conv. and Sel. are abbreviations for ethanol conversion and acetaldehyde selectivity, respectively.

ethanol dehydrogenation activity can be attributed to the reconstruction of the surface structure of metallic copper (Cu<sup>0</sup>) nanoparticles, without altering their size, dispersion, or oxidation state. Meanwhile, it was

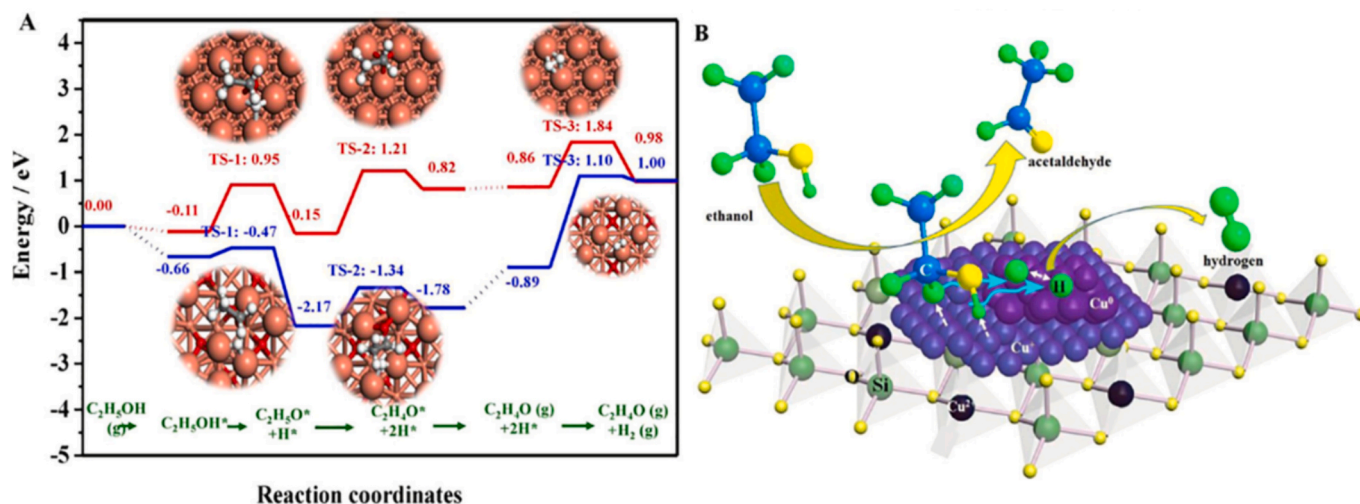
proven that the stepped and defect-rich Cu(211) surface exhibits higher activity than the relatively flat Cu(111) surface. As shown in Fig. 1, (a) illustrates the reaction mechanism for ethanol dehydrogenation to acetaldehyde on Cu, where the rate-determining step is CH<sub>3</sub>CH<sub>2</sub>O\* → CH<sub>3</sub>CHO\* + H. (b) DFT calculations reveal that the Cu(211) surface exhibits stronger adsorption of the ethoxy intermediate (CH<sub>3</sub>CH<sub>2</sub>O) and a significantly lower activation energy barrier for the rate-determining C–H bond cleavage compared to the Cu(111) surface. In contrast, Lin et al. [14] demonstrated that Cu<sup>+</sup> species are the active sites in Cu-based catalysts for the dehydrogenation of ethanol to acetaldehyde, which can efficiently catalyze the cleavage of C–H and O–H bonds in ethanol and facilitate its adsorption and activation.

Some studies indicate that during ethanol dehydrogenation, Cu-based catalysts show a synergistic mechanism between Cu<sup>+</sup> and Cu<sup>0</sup> species [28,29,51]. The study by Pang et al. [29] demonstrated via DRIFTS experiments that ethanol molecules preferentially and strongly adsorb onto Cu<sup>+</sup> sites rather than Cu<sup>0</sup>. As shown in Fig. 2, DFT calculations revealed that the energy barriers for both O–H and C–H bond cleavages are significantly lower on Cu<sup>+</sup> sites (Cu<sub>2</sub>O (100) surface) than on Cu<sup>0</sup> sites (Cu (111) surface), with values of 0.19 eV versus 1.06 eV for O–H cleavage and 0.83 eV versus 1.36 eV for C–H cleavage. In contrast, the energy barrier for H<sub>2</sub> formation is much higher on Cu<sup>+</sup> sites than on Cu<sup>0</sup> sites (1.99 eV vs. 0.98 eV). The complementary functions of Cu<sup>+</sup> and Cu<sup>0</sup> in the reaction are evident. Cu<sup>+</sup>, characterized by a d<sup>10</sup> electronic configuration, exhibits strong Lewis acidity, enabling strong interactions with oxygen-containing species such as the oxygen atom of ethanol. This electronic structure allows Cu<sup>+</sup> to exhibit high selectivity in activating both the O–H and C–H bonds of ethanol. In contrast, Cu<sup>0</sup> possesses metallic character with a d<sup>10</sup>s<sup>1</sup> electronic configuration, offering good electrical conductivity and high capability for hydrogen atom adsorption. As a result, Cu<sup>0</sup> sites are more effective for hydrogen atom combination and H<sub>2</sub> release.

It is not difficult to imagine that the different proportions of Cu<sup>0</sup> and Cu<sup>+</sup> in Cu-based catalysts would affect this synergistic effect, leading to different catalytic activities. Recently, Chao Tian et al. [27] found that the acetaldehyde productivity on their developed Cu@mSiO<sub>2</sub> catalyst



**Fig. 1.** (a) Proposed reaction mechanism for ethanol dehydrogenation on Cu surfaces. (b) Gibbs free energy diagram obtained from DFT calculations for ethanol dehydrogenation on Cu (111) (black) and Cu (211) (red). All energies are relative to an ethanol molecule in the gas phase and a clean Cu surface. Intermediates adsorbed on Cu surfaces are marked with an asterisk. Source: Adapted from ref [23] with permission. Copyright 2019, published American Chemical Society.



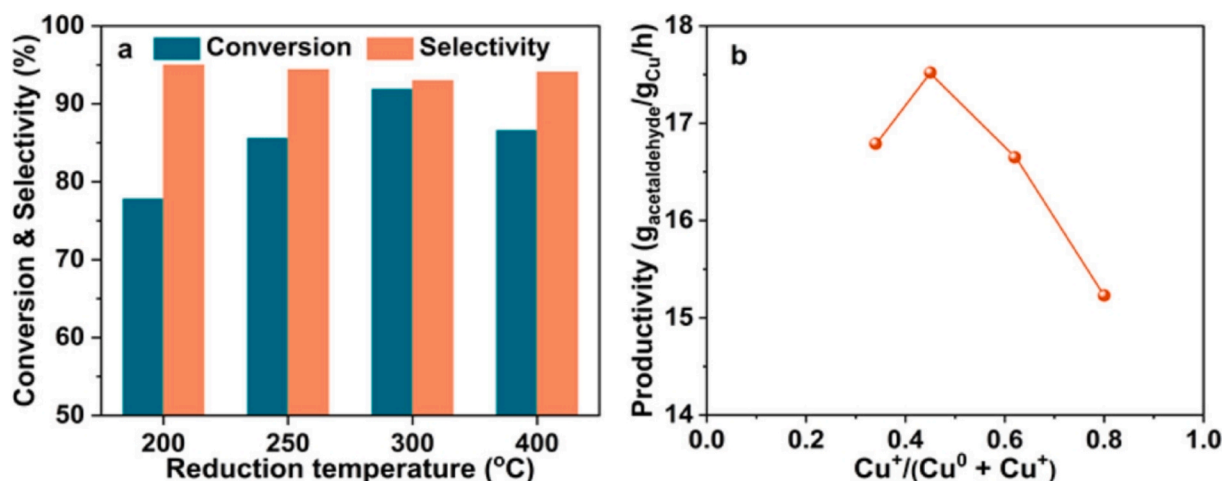
**Fig. 2.** Profiles for dehydrogenation of ethanol to acetaldehyde. (A) Energy profile for ethanol dehydrogenation on Cu (111) (red) and Cu<sub>2</sub>O (100) (blue) and (B) A schematic illustration of ethanol dehydrogenation on the Cu-MFI-AE catalyst interface. All energies are relative to an ethanol molecule in the gas phase and a clean Cu or Cu<sub>2</sub>O surface. Intermediates adsorbed on surface are marked with an asterisk. Source: Adapted from ref [29] with permission. Copyright 2020, published American Chemical Society.

can be controlled by adjusting the Cu<sup>+</sup>/(Cu<sup>0</sup> + Cu<sup>+</sup>) ratio via varying the reduction temperature, as illustrated in Fig. 3.

In contrast to the widely reported Cu<sup>+</sup> and Cu<sup>0</sup> species as the active sites in copper catalysts, Li et al. [47] revealed that highly dispersed Cu<sup>δ+</sup> (1 < δ < 2) species on the MFI support serve as the essential active sites for ethanol dehydrogenation, with the δ value quantitatively determined to be ≈1.38 by XANES. This species is not a simple physical mixture of Cu<sup>+</sup> and Cu<sup>0</sup>, but rather a distinct and stable single interfacial site with electronically modulated properties. Researchers' understanding of the catalytic process becomes progressively deeper. Future research trends could focus more on the precise control and stabilization of these highly active intermediate-valence species through catalyst design, as well as gaining an in-depth understanding of their dynamic evolution under reaction conditions.

Cu catalysts exhibit high selectivity toward acetaldehyde. However,

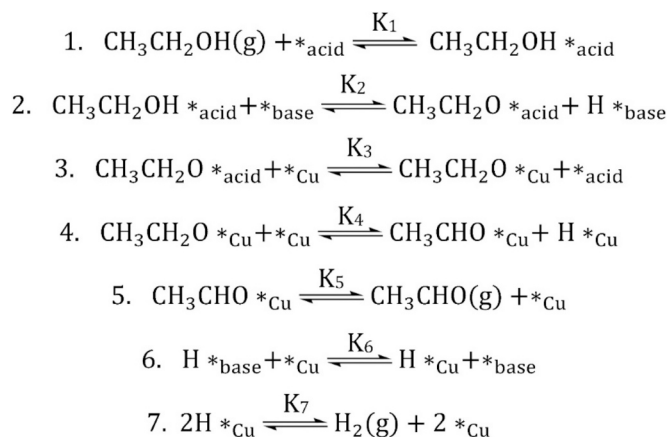
the reaction mechanism, the rate-limiting step, and the influence of support acidity on the reaction remain incompletely understood. The study by Hanukovich et al. [52] revealed that the critical role played by the acid sites of metal oxide supports is strongly dependent on the ethanol partial pressure. At low ethanol pressure (~1 mbar), the reaction proceeds predominantly on the Cu surface with O–H bond cleavage as the rate-limiting step, and the support exerts minimal influence. In contrast, at high ethanol pressure (~70 mbar), the Cu surface becomes poisoned by reactive intermediates, and the reaction shifts to the Cu–support interface. Under these conditions, supports with strong Lewis acidity (e.g., TiO<sub>2</sub> and Al–ZrO<sub>2</sub>) enhance the reaction rate by over an order of magnitude compared to weakly acidic ZrO<sub>2</sub> via a cooperative mechanism—wherein the Lewis acid sites facilitate ethanol O–H bond cleavage, and adjacent interfacial Cu sites are responsible for the subsequent α-C–H bond cleavage as the rate-limiting step, thereby



**Fig. 3.** (a) Dehydrogenation of ethanol over the 7Cu@mSiO<sub>2</sub> catalyst reduced in 10 % H<sub>2</sub>/Ar at different temperatures. (b) Acetaldehyde productivity as a function of the Cu<sup>+</sup> / (Cu<sup>0</sup> + Cu<sup>+</sup>) ratio. Reaction conditions: 260 °C, WHSV = 1.5 h<sup>-1</sup>, after 1 h of the reaction. Source: Adapted from ref [27] with permission. Copyright 2025, published American Chemical Society.

significantly boosting the catalytic efficiency. Based on their analysis, they proposed an interfacial reaction mechanism, as depicted in Scheme 2, whereby Cu and the support act synergistically under the combined influence of high ethanol pressure and a strongly acidic support to significantly enhance the catalytic efficiency.

Under ambient pressure, ethanol conversion primarily follows two competing pathways: dehydrogenation and dehydration, producing acetaldehyde and ethylene, respectively. The dehydration reactions readily occur at the Brønsted acid sites of the catalyst [34]. Whereas, as previously mentioned, the dehydrogenation pathways depend on metal active sites coupled with an appropriate amount of Lewis acid sites (such as the synergistic interaction between Cu<sup>0</sup> and Cu<sup>+</sup>). Using a support material with neutral properties presents a viable strategy to address this challenge. As reported by Liu et al. [24], the use of neutral SiO<sub>2</sub> as a support not only effectively suppresses the formation of acidic sites, significantly reducing dehydration side reactions, but also directs the reaction toward the highly selective dehydrogenation pathway, thereby enabling the efficient production of acetaldehyde. Furthermore, Liu et al. [34] treated Beta zeolite via “dealumination”, transforming it from a carrier rich in Brønsted acid (which promotes dehydration) into a purified support (Si-Beta) nearly free of Brønsted acid but rich in silanol nests, thereby suppressing the ethanol dehydration reaction.



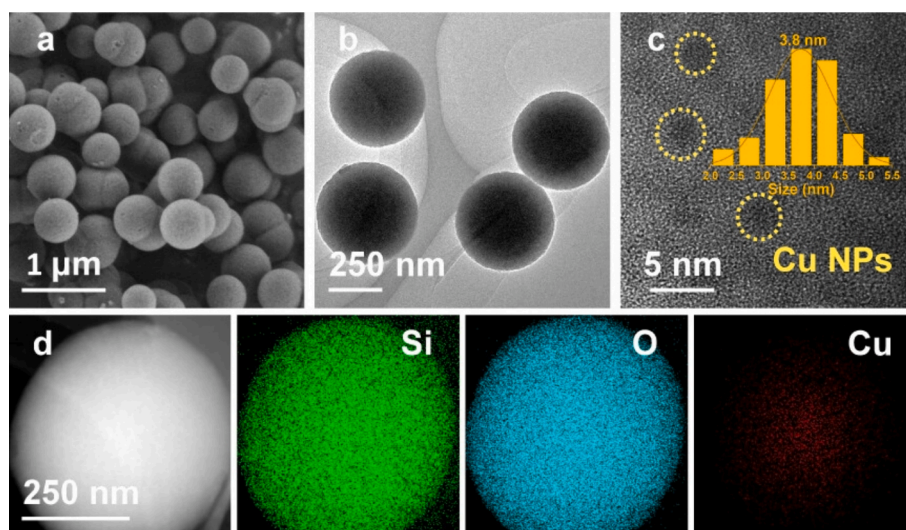
**Scheme 2.** Reaction Mechanism of Ethanol Dehydrogenation over Cu/Support<sup>a</sup>. <sup>a</sup>where \*<sub>Cu</sub> represents Cu adsorption sites, \*<sub>base</sub> represents support basic sites, and \*<sub>acid</sub> represents Lewis acid sites. Source: Adapted from ref [52] with permission. Copyright 2019, published American Chemical Society.

## 2.2. Stability strategies for Cu-based catalysts

Cu-based catalysts for ethanol dehydrogenation to acetaldehyde face a critical challenge: achieving high activity and selectivity requires Cu species with small particle size (<10 nm), high dispersion (20–60 %), and an optimized Cu<sup>+</sup>/Cu<sup>0</sup> ratio to fully expose highly active sites [38]. However, the low Tammann temperature of copper (405 °C) makes it prone to migration, agglomeration, and Ostwald ripening during reaction, ultimately leading to sintering and deactivation [24,53]. In addition, although common oxide supports such as SiO<sub>2</sub>, Al<sub>2</sub>O<sub>3</sub>, and ZrO<sub>2</sub> have high surface areas and excellent thermal stability, their surface chemical properties (e.g., acid-base and hydroxyl density) and pore structure [54] tend to trigger secondary reactions, that is, aldolization and ketonization [10,55], which can reduce acetaldehyde selectivity.

To address these challenges, some researchers have adopted a “physical confinement” strategy, which involves using nanoscale spatial structures to physically isolate copper active sites, preventing their migration and agglomeration, thereby enabling them to participate in catalysis in a stable and efficient manner. For example, Lin et al. [14] used an in situ method to encapsulate ultra-small Cu metal nanoclusters (~1.8 nm) in zeolite (Silicalite-1), resulting in a Cu@S-1 catalyst with excellent stability. The spatially confined environment of the zeolite (Silicalite-1) not only effectively restricts the movement of Cu entities, thus preventing the Cu-metal sintering during the catalytic process, but also induces a large number of well-retained Cu<sup>+</sup> species (up to ~70 %). Recently, Tian et al. [27] prepared a core-shell-structured Cu@mSiO<sub>2</sub> catalyst, in which copper nanoparticles are encapsulated in a mesoporous silica shell, using a modified Stöber method. The geometrical confinement effect of the mesoporous silica shells effectively prevented metal sintering during the reaction, thus significantly improving the stability and activity of the catalyst. As shown in Fig. 4, the reduced catalyst (7Cu@mSiO<sub>2</sub>) exhibits uniform spherical particles. Meanwhile, TEM images reveal that the core of the 7Cu@mSiO<sub>2</sub> catalyst consists of highly dispersed copper oxide nanoparticles with an average size of 3.8 nm.

In addition, Pang et al. [29] developed a hierarchical echinus-like Cu-MFI catalyst by leveraging the spatial confinement effect. The formation of Cu–O–Si on the interface between Cu and the hierarchical MFI effectively prevented the aggregation of Cu nanoparticles and stabilized Cu<sup>+</sup> species. Liu et al. [24] successfully constructed a unique hollow octahedral-structured Cu/ZnO@SiO<sub>2</sub> catalyst by introducing silica (SiO<sub>2</sub>) in situ into a CuZn-BTC metal-organic framework (MOF) precursor via a steam-assisted hydrolysis strategy. The SiO<sub>2</sub> shell not only acts as a physical barrier to effectively suppress the migration and

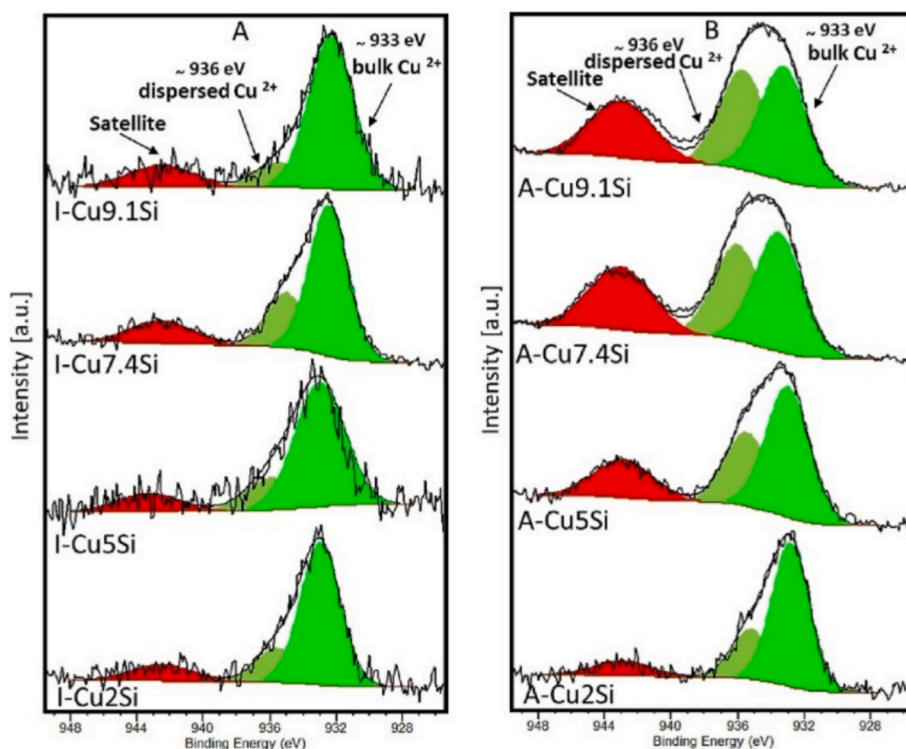


**Fig. 4.** (a) SEM image, (b) low and (c) high-magnification TEM images, and (d) HAADF STEM mapping images of reduced 7Cu@mSiO<sub>2</sub>. The inset of Fig. 1c is the Cu NPs size distribution. Source: Adapted from ref [27] with permission. Copyright 2025, published American Chemical Society.

agglomeration of Cu nanoparticles at high temperatures, but its mild decomposition process also avoids violent exothermic reactions during the calcination of the MOF precursor, thereby achieving highly dispersed and stably anchored Cu and ZnO nanoparticles. Ultimately, the catalyst demonstrates exceptional stability in the ethanol dehydrogenation reaction, maintaining performance for 200 h at 350 °C.

Using classical catalyst preparation methods (e.g., impregnation), it is a challenge to obtain a stable and homogeneous dispersion of Cu over SiO<sub>2</sub> owing to the weak support-active phase interactions. Some other researchers have employed a “chemical anchoring” strategy to enhance metal-support interactions through unique synthetic approaches. For example, Guo et al. [50] selected low-cost ethylenediamine and

propylenediamine as effective chelators agents to stabilize the Cu species under basic conditions, and then anchored them into the framework of MFI (silicate-1) in hydrothermal process. The obtained Cu@MFI-R catalysts showed very high ethanol conversion (>96 %) and acetaldehyde selectivity (98 %) over 300 h of time on stream under the reaction conditions of 250 °C, a WHSV of 0.55 h<sup>-1</sup>, and nitrogen as carrier gas (40 mL/min). Pampararo et al. [56] successfully prepared mesoporous Cu-SiO<sub>2</sub> catalysts with strong metal-support interaction (SMSI) via an aerosol-assisted sol-gel method. This interaction enabled the high dispersion of copper species at the nanoscale (~10 nm) and their firm anchoring within the silica matrix, effectively suppressing the sintering of copper particles during reaction and ensuring structural stability of



**Fig. 5.** High-resolution XPS spectra of Cu 2p<sub>3/2</sub> on fresh I-catalysts (A) and A-catalysts (B). Source: Adapted from ref [56] with permission. Copyright 2023, published Elsevier.

the catalyst. As shown in Fig. 5, the prominent Cu 2p<sub>3/2</sub> XPS peak at approximately 936 eV in the aerosol-prepared catalyst (B) indicates well-dispersed Cu(II) interacting with the silica network. This may suggest the presence of  $\text{—Cu—O—Si—}$  species (i.e., well dispersed CuO and or copper phyllosilicate-like species). It was found that deactivation during an 8-hour stability test at 300 °C primarily originated from reversible coke deposition. As a result, a simple mild oxidative treatment at 515 °C efficiently removed the carbon deposits, nearly fully restoring the catalytic activity and demonstrating excellent regenerability.

Additionally, Ausavasukhi et al. [57] explored the effects of copper incorporated hydrotalcite-derived oxides at different heat treatment temperatures on the catalyst structure and performance. The study found that the catalyst (c-10Cu/HT600) treated at 600 °C generated  $\text{CuAl}_2\text{O}_4$  spinel structure linked to the copper oxide species, which can suppress copper agglomeration, leading to the catalytic stability.

Compared to  $\text{SiO}_2$ ,  $\text{Al}_2\text{O}_3$ , and  $\text{ZrO}_2$ , carbon materials possess a chemically inert nature and hydrophobic properties, which can effectively inhibit secondary reactions and improve acetaldehyde selectivity. For example, Wang et al. [49] demonstrated the superior performance of a Cu catalyst supported on mesoporous carbon (Cu/MC) over its silica-based counterpart (Cu/SBA-15). Under identical conditions (280 °C, GHSV = 8600 h<sup>-1</sup>), Cu/MC achieved a high acetaldehyde selectivity of 94.0 % and a space-time yield (STY) of 225 h<sup>-1</sup>, doubling that of Cu/SBA-15 (STY = 112 h<sup>-1</sup>). This exceptional performance was attributed to the unique advantages of the mesoporous carbon support, specifically its strong enrichment capability for ethanol molecules which enhances the reaction rate, and its chemically inert surface which lacks acidic/basic sites, thereby effectively suppressing acetaldehyde side reactions.

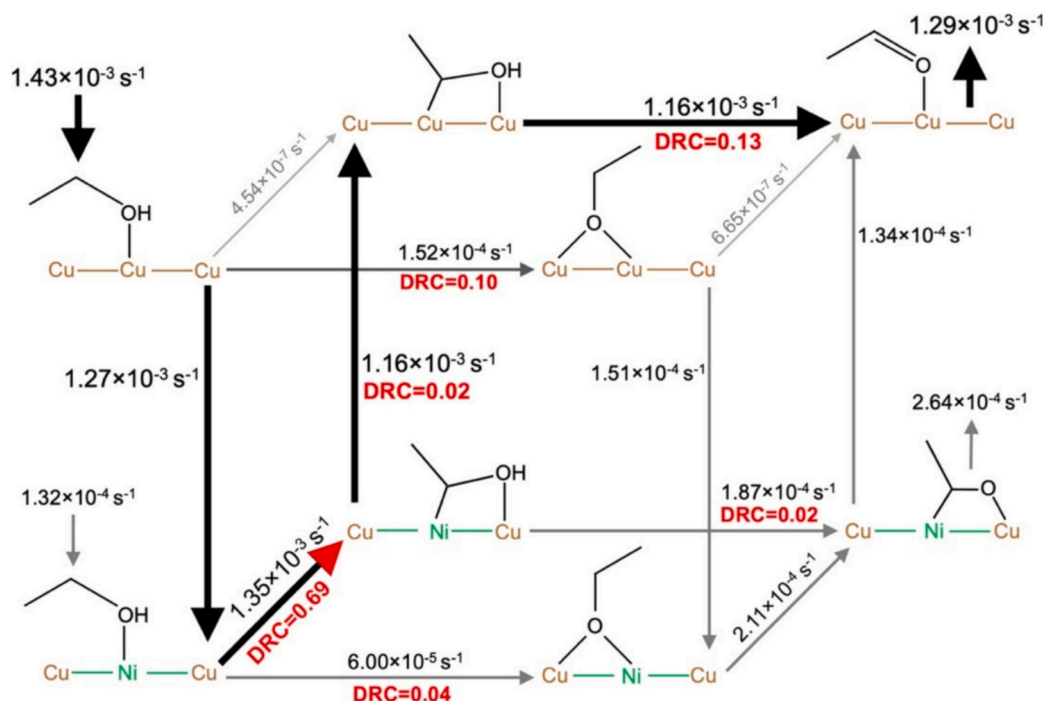
In addition, doping various metal promoters into Cu-based catalysts to form bimetallic catalysts can effectively regulate product selectivity, alter reaction pathways, and significantly enhance catalyst stability. Samira Amokrane et al. [26] added transition metals (Ag, Co, Ni, Cd and

Pt) to copper to investigate the ethanol dehydrogenation process on a copper bimetallic systems. Monometallic copper has been shown to agglomerate and deactivate, which can be prevented by adding Ag, Co, Cd, or Ni. Among them, Ni-Cu improves conversion but reduces acetaldehyde selectivity by producing the byproduct  $\text{CH}_4$ . Ag reacts weakly with ethanol, resulting in decreased conversion. Pt changes the chemical route, causing the process to yield diethyl ether. It is worth noting that Cd-Cu greatly increases the sintering resistance of the catalysts, increasing their stability.

### 2.3. Single-atom alloy catalysts

Single-atom alloy catalysts exhibit outstanding activity and selectivity in the dehydrogenation of ethanol to acetaldehyde. Studies indicate that single-atom alloys such as NiCu, NiAu, and Pt-Cu achieve efficient activation of the C-H and O-H bonds in ethanol molecules through synergistic interactions between metal sites, following a “dual-site synergistic mechanism.” This approach effectively suppresses side reactions like C-C bond cleavage [25,58,59].

Patel et al. [58] elucidated the microscopic mechanism of the non-oxidative ethanol dehydrogenation reaction on NiCu single-atom alloys by integrating kinetic experiments, in situ infrared spectroscopy, surface science model studies, DFT calculations, and microkinetic simulations. The research reveals that the reaction follows a dual-site cooperative mechanism: a single Ni atom is responsible for ethanol activation and the cleavage of its  $\alpha\text{-C-H}$  bond (the rate-determining step), generating the key intermediate 1-hydroxyethyl ( $\text{CH}_3\text{CHOH}^*$ ), which subsequently spills over to adjacent Cu sites where the O-H bond is cleaved to ultimately form acetaldehyde. Microkinetic simulation results (Fig. 6) confirm that this “Ni-catalyzed C-H cleavage, Cu-catalyzed O-H cleavage” pathway is the dominant reaction channel, accounting for approximately 74 % of the acetaldehyde yield. Density



**Fig. 6.** Steady-state reaction rates (black text on arrows) of elementary steps in ethanol dehydrogenation over NiCu(111) and the DRC (red text on arrows) of important transition states (TS). The thickness of the arrows reflects the rate of the elementary steps, while the step containing the TS with the highest DRC,  $\alpha\text{-C}$  dehydrogenation of ethanol on Ni, is marked with a red tip. In the dominant reaction pathway, ethanol first adsorbs on a free Cu site and diffuses to Ni. The Ni site then cleaves the adsorb ethanol's  $\alpha\text{-C-H}$  bond to form the hydroxyethyl intermediate. The hydroxyethyl intermediate then diffuses back to a free Cu site, where its O-H bond is cleaved to form acetaldehyde, which desorbs to close the catalytic cycle. Importantly, the dominant pathway is facilitated by the spillover of intermediates between Cu and Ni sites. Conditions are:  $T = 180$  °C,  $P(\text{EtOH}) = 0.05$  kPa,  $P(\text{H}_2) = 10^{-6}$  kPa and  $P(\text{CH}_3\text{CHO}) = 10^{-6}$  kPa. Source: Adapted from ref [58] with permission. Copyright 2023, published American Chemical Society.

functional theory reveals that Ni doping in Cu single-atom alloys (SAAs) significantly lowers the C–H bond scission barrier from 1.51 eV on pure Cu to 1.21 eV, indicating strong electronic modulation by isolated Ni atoms. When Ni coverage remains below 0.03 ML, Ni disperses atomically, preventing ensemble formation and suppressing side reactions such as C–C cleavage and coking. Experiments showed that the NiCu SAA exhibited a 12-fold higher acetaldehyde yield, with the peak desorption temperature of acetaldehyde lowered by 30 K (from 320 K to 290 K) compared to pure Cu.

Furthermore, Giannakakis et al. [25] investigated the geometric and electronic structure of the active sites and the reaction mechanism of NiAu single-atom alloy (SAA) catalysts. Owing to the electronegativity difference between Au and Ni, the isolated Ni atoms carry a partial positive charge. This unique electronic structure not only enhances the ability of Ni to activate both C–H and O–H bonds in ethanol but also preserves high selectivity. Spectroscopic and theoretical studies revealed that the Ni sites undergo a reversible oxidation state cycle during reaction, which prevents strong over-binding of intermediates and thus suppresses unselective C–C scission. Moreover, the atomic dispersion of Ni, which was maintained at a surface coverage below 0.15 monolayer (ML) on model surfaces and an overall Ni/Au atomic ratio below 1 % in nanoparticles, effectively suppresses the formation of Ni ensembles, thereby eliminating undesirable side reactions such as C–C bond cleavage.

Wang et al. [59] formed a Pt–Cu single atom alloy (SAA) by adding a small amount of Pt to the surface of Cu(111), which resulted in a sixfold enhancement of the ethanol dehydrogenation activity of Cu(111). The study demonstrated that Pt–Cu SAAs are much more active than Cu(111) for converting ethanol to acetaldehyde and hydrogen in the absence of surface oxygen and water. Specifically, below 160 K, the Pt active site effectively activates the O–H bond of ethanol, and the generated ethoxylate spillover to the Cu sites, which activates the C–H bond at 310 K, ultimately generating acetaldehyde desorption from Cu(111). This metal synergy at the single-atom level improves the catalytic activity while retaining the high selectivity of Cu.

#### 2.4. Metal oxide catalysts

In the non-oxidative dehydrogenation of ethanol to acetaldehyde, metal oxide catalysts possess distinctive active sites and varied reaction mechanisms.

In the study by Raynes and Taylor [60], a catalyst comprising zinc oxide on mordenite (ZnO/M-MOR) was prepared via wetness impregnation. The catalytic system functioned synergistically: ZnO facilitated the C–H bond cleavage essential for ethanol dehydrogenation, while the  $\text{Rb}^+$  cations exchanged into the mordenite framework optimized the surface acidity, thereby minimizing the ethylene formation pathway and achieving a selectivity of less than 0.9 %. Concurrently, the zeolitic pore structure ensured a high dispersion of the active ZnO phase, leading to an outstanding acetaldehyde productivity of  $48 \text{ mmol mmol}_{\text{Zn}}^{-1} \text{ h}^{-1}$  and remarkable stability against carbon deposition during a 120-hour longevity test.

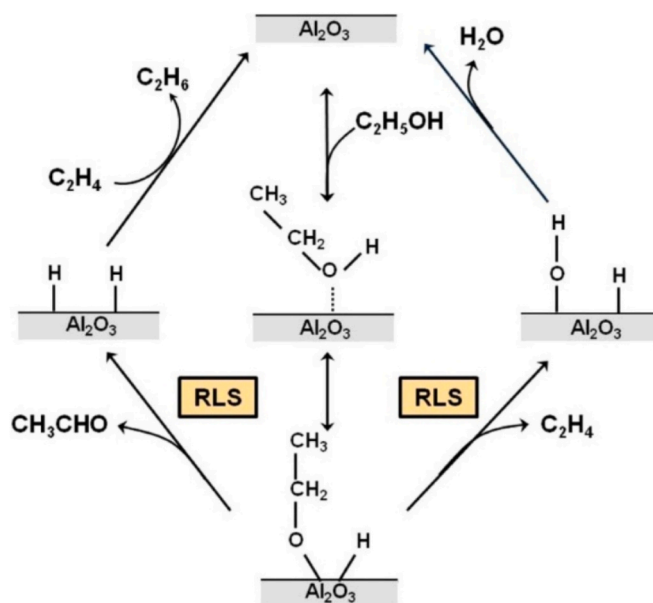
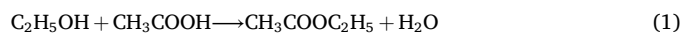
Furthermore, Autthanit et al. [61] prepared VOx/SBA-15 catalysts using the hydrothermal method and investigated their performance in the non-oxidative dehydrogenation of ethanol. The study revealed that isolated monomeric  $\text{V}^{4+}$  species, together with weak basic sites on the catalyst surface, constituted key components of the catalytic active centers. These sites were crucial for promoting reaction steps involving C–H bond cleavage and hydrogen transfer. Among these catalysts, V–Zr–La/SBA-15-HT exhibited a high concentration of monomeric VOx species at 400 °C, achieving an ethanol conversion of 80 % with 48 % acetaldehyde selectivity, resulting in an acetaldehyde yield of approximately 38 %.

DeWilde et al. [62] elucidated the reaction mechanism of ethanol dehydration and dehydrogenation on a  $\gamma\text{-Al}_2\text{O}_3$  catalyst at high temperatures (>350 °C). The study ruled out the direct dehydrogenation

pathway of ethanol to form acetaldehyde and hydrogen gas, instead proposing a key “indirect hydrogen transfer” mechanism (Scheme 3): The cleavage of the  $\text{C}_\alpha\text{--H}$  bond in an adsorbed surface ethoxy intermediate is the rate-determining step for acetaldehyde formation. The resulting surface hydrogen atoms do not combine to form gaseous  $\text{H}_2$  but instead directly transfer to another ethylene molecule, hydrogenating it to ethane. This mechanism perfectly explains the experimentally observed 1:1 co-production of acetaldehyde and ethane and reveals that this dehydrogenation pathway shares a common surface ethoxy precursor with the ethylene dehydration pathway. The selectivity between the two products is determined by the relative rates of  $\text{C}_\alpha\text{--H}$  versus  $\text{C}_\beta\text{--H}$  bond cleavage.

### 3. Dehydrogenative coupling of ethanol to ethyl acetate

Ethyl acetate is a widely used low-toxicity solvent and a key chemical intermediate. There are three main industrial production routes for ethyl acetate. As shown in Eq. (1), the traditional production method is the Fischer esterification process, in which ethanol and acetic acid react under the catalysis of strong acids such as sulfuric acid or p-toluene-sulfonic acid [65,66]. Eq. (2) is produced by the Tishchenko process, popular in Germany and Japan, which uses a solid base catalyst (e.g., alumina, sodium oxide, or alkaline earth oxide) to catalyze the dimerization of acetaldehyde to form ethyl acetate [67–69]. Eq. (3), which has been commercialized, uses modified clays or heteropolyacids as catalysts to generate ethyl acetate by the addition reaction of acetic acid and ethylene [70–72]. However, as shown in Eq. (4), ethanol dehydrogenation coupling to ethyl acetate is more attractive than the production methods described above (laboratory conditions). This method is simple and efficient, avoids highly toxic or corrosive catalysts, and produces hydrogen as a by-product, and has therefore received much attention in recent years [73,74]. This review systematically analyzes the catalytic performance of various catalysts for the dehydrogenation coupling of ethanol to ethyl acetate, as summarized in Table 3.



**Scheme 3.** Proposed Mechanism for the Combined Synthesis Pathways of Ethylene and Acetaldehyde from a Common Ethoxy Surface Intermediate on  $\gamma\text{-Al}_2\text{O}_3$ . Source: Adapted from ref [62] with permission. Copyright 2014, published American Chemical Society.

**Table 3**  
Catalysts for dehydrogenation of ethanol to ethyl acetate.<sup>a</sup>

| Catalyst                                  | Reaction conditions                                                  | Conv.  | Sel.   | Feature                                                                                                                                     | Ref. |
|-------------------------------------------|----------------------------------------------------------------------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------|------|
| 50 Cu-Cr<br>(50 wt% CuO)                  | N <sub>2</sub> , WHSV = 3h <sup>-1</sup> , 300 °C, 1 bar, reduce 2 h | 92.8 % | 5 %    | Cu <sup>0</sup> is responsible for ethanol dehydrogenation, Cu <sup>+</sup> and CuCr <sub>2</sub> O <sub>4</sub> provide Lewis acid sites   | [32] |
| CuCeZr<br>(10 wt% Cu, 10 wt% Ce)          | N <sub>2</sub> , WHSV = 1h <sup>-1</sup> , 220 °C, 1 MPa             | 55.9 % | 87.5 % | Ce doping enhances Cu <sup>+</sup> species and suppresses strong acid-base sites, maintaining stability over 1000 h of continuous reaction. | [78] |
| CuCeZrO<br>(7 wt% Cu, 63 wt% Zr)          | WHSV = 1h <sup>-1</sup> , 240 °C, 1.0 MPa                            | 40 %   | 71 %   | m-ZrO <sub>2</sub> has a low density of strong basic sites, inhibiting acetaldehyde condensation; also has higher oxygen mobility.          | [80] |
| CCZM-5 %<br>(5.8 wt% Cu, 11.9 wt% Zr)     | LHSV = 1h <sup>-1</sup> , 220 °C, 0.1 MPa,                           | ~45 %  | ~75 %  | Mn modification enhances thermal stability, with almost no carbon deposition after 140 h of reaction.                                       | [79] |
| Cu-Zn-Zr-Al-O<br>(Cu/Zn/Zr/Al = 12:1:2:4) | N <sub>2</sub> , WHSV = 0.53h <sup>-1</sup> , 220 °C, 0.8 MPa        | 51.4 % | 92.9 % | High pressure reduces by-product formation, enhancing selectivity.                                                                          | [21] |
| Pd-Zn-CMR                                 | catalytic membrane reactor, 250 °C, 1 MPa, Contact time: 84 s        | 34 %   | 40.1 % | A longer contact time promotes the coupling of the intermediate acetaldehyde, thereby significantly enhancing the ethyl acetate yield.      | [83] |

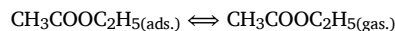
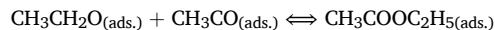
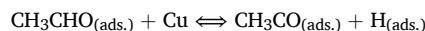
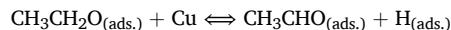
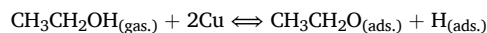
<sup>a</sup> Reactions were performed in a fixed bed reactor if not specified; Conv. and Sel. are abbreviations for ethanol conversion and ethyl acetate selectivity, respectively.



### 3.1. Cu-based catalysts

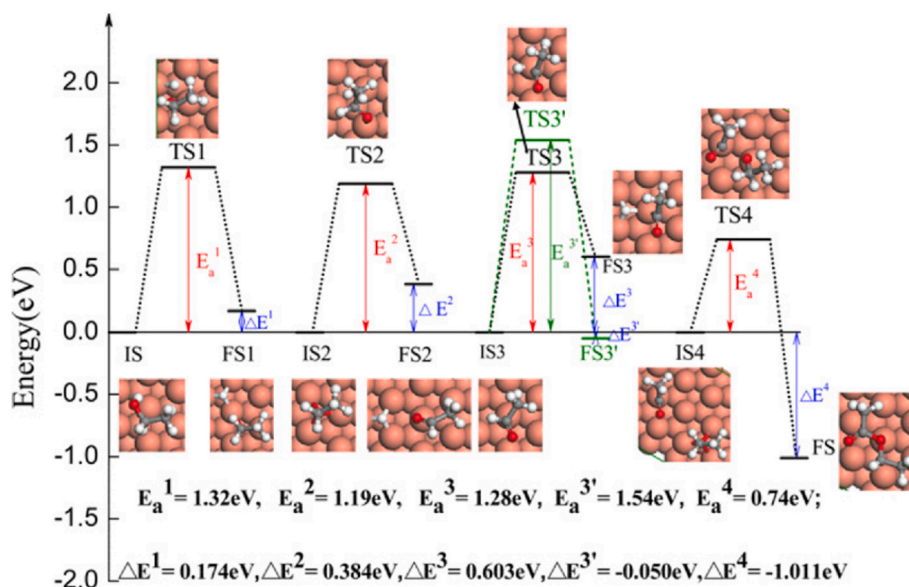
Cu-based catalysts have shown excellent performance in the dehydrogenation coupling of ethanol to ethyl acetate, and there are two explanations for their reaction pathways. One was proposed by Inui et al. [21], who, using a Cu-Zn-Zr-Al-O catalyst under pressurized conditions (0.1–1.0 MPa, 200–260 °C), outlined a three-step mechanism: (1) ethanol is firstly dehydrogenated to acetaldehyde; (2) subsequently, acetaldehyde is coupled with another ethanol to form hemiacetal; and (3) finally, hemiacetal is further dehydrogenated to ethyl acetate. Another reaction pathway proposed was by Colley et al. [75] through temperature-dependent adsorption studies (TDSS) and temperature-programmed reaction (TPR) experiments over Cu/Cr<sub>2</sub>O<sub>3</sub> catalyst. Their mechanism involved four key steps: (1) ethanol as ethoxy adsorbed on the Cu active site; (2) the adsorbed ethoxy species was dehydrogenated to form an acetyl species; (3) the adsorbed ethoxy species reacted with the adsorbed acetyl species to form adsorbed ethyl ethanoate; and (4) migration and strong adsorption of the ethyl acetate

product onto a Brønsted acid site on the Cr<sub>2</sub>O<sub>3</sub> support, from which its desorption at 407 °C was identified as the rate-determining step. The specific reaction process is as follows:



Li et al. [73] later investigated the reaction mechanism of ethanol dehydrogenation to ethyl acetate on the Cu(111) surface by density functional theory (DFT) calculations, focusing on comparing the two possible reaction pathways mentioned above. The results show that the pathway proposed by Colley is more kinetically favorable. In the Colley pathway [75], the activation energy for the dehydrogenation of ethanol to form an ethoxy group and hydrogen is 1.32 eV, while the activation energy for the subsequent dehydrogenation of ethoxy to acetaldehyde and hydrogen is 1.19 eV. The activation energy for the dehydrogenation of acetaldehyde to acetyl and hydrogen is 1.28 eV, and the activation energy for the coupling of acetyl and ethoxy to form ethyl acetate is as low as 0.74 eV. (Fig. 7) In contrast, in the pathway proposed by Inui [21], the activation energy for the nucleophilic addition of ethanol and acetaldehyde to form a hemiacetal intermediate is as high as 1.54 eV—substantially higher than any step in the Colley pathway. This significant difference in activation energies highlights the greater kinetic advantage of Colley's pathway on the Cu(111) surface, making it more likely to dominate the formation of ethyl acetate. Furthermore, the research work published by Santacesaria et al. [74] systematically elucidated that the efficient one-step dehydrogenation of ethanol to ethyl acetate can be achieved by employing an optimized copper-chromium catalyst (BASF Cu-1234) under suitable reaction conditions. The study revealed that at moderate pressure (~20 bar), the reaction pathway shifts from favoring acetaldehyde formation at low pressure to efficiently generating ethyl acetate. The essence of this phenomenon is that elevated pressure increases the concentration of reactive intermediates on the catalyst surface. The catalyst's unique bifunctional active center—comprising metallic copper (Cu<sup>0</sup>) responsible for dehydrogenation and Brønsted acid sites originating from the reduced copper chromite spinel, working in synergy—selectively directs the intermediates toward ethyl acetate synthesis via the hemiacetal pathway while effectively suppressing side reactions such as acetaldehyde self-condensation. Evidently, this work also supports the mechanism proposed by Colley et al.

The study by Zhang et al. [30] revealed a well-defined site-specific mechanism on the reduced Cu-Cr catalyst surface, which consists of Cu<sup>0</sup>, Cr<sub>2</sub>O<sub>3</sub>, and CuCr<sub>2</sub>O<sub>4</sub>: the ethanol dehydrogenation step occurs on metallic sites (Cu<sup>0</sup>), while the CuCr<sub>2</sub>O<sub>4</sub> phase acts as a structural promoter to stabilize these metallic sites. Through programmed temperature desorption (EA-TPD) experiments, they demonstrated that the ethyl acetate product hardly adsorbs on pure Cu<sup>0</sup> sites or the Brønsted acid sites of the catalyst, but instead adsorbs on the Lewis acid sites provided by the Cr<sub>2</sub>O<sub>3</sub> phase. These Lewis acid sites promote the desorption of ethyl acetate, thereby shifting the reaction equilibrium forward and effectively suppressing side reactions such as the condensation of acetaldehyde intermediates to form butanone. Wang et al. [31] found that in Cu/ZrO<sub>2</sub> catalysts (Cu/Zr = 1:1) prepared by co-precipitation, the synergistic effect between Cu<sup>+</sup> and Cu<sup>0</sup> species significantly enhanced the catalytic performance for ethanol dehydrogenation to ethyl acetate. The presence of Cu<sup>+</sup> provides additional Lewis acid sites for acetaldehyde activation, while its interaction with Cu<sup>0</sup> collectively promotes the efficient conversion of ethanol. Preedavijitkul et al. [32] further investigated the effects of different Cu/Cr molar ratios on the ethanol dehydrogenation reaction. They found that at a molar ratio of 0.98 and under



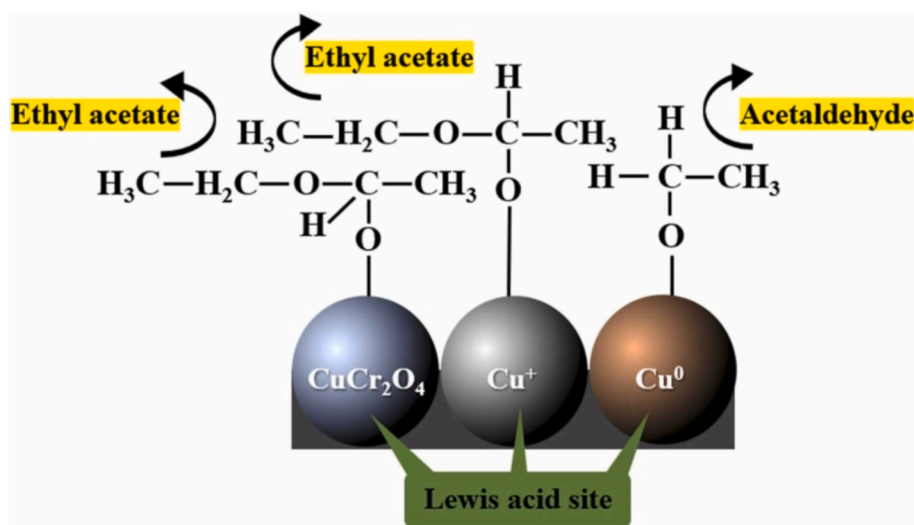
**Fig. 7.** Schematic energetic diagram for reaction  $2\text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_3\text{COOC}_2\text{H}_5 + 2\text{H}_2$  on Cu (111) surface, going through the IS( $\text{C}_2\text{H}_5\text{OH}$ ), FS1( $\text{C}_2\text{H}_5\text{O} + \text{H}$ ), IS2( $\text{C}_2\text{H}_5\text{O}$ , FS2( $\text{CH}_3\text{CHO} + \text{H}$ ), IS3( $\text{CH}_3\text{CHO}$ , FS3( $\text{CH}_3\text{CO} + \text{H}$ ), FS3'( $\text{CH}_3\text{CH}(\text{OH})\text{OC}_2\text{H}_5$ ), IS4( $\text{CH}_3\text{CO} + \text{C}_2\text{H}_5\text{O}$ ), and FS( $\text{CH}_3\text{COOC}_2\text{H}_5$ ). Graphics of IS, TS1, FS1, IS2, TS2, FS2, IS3, TS3, FS3, IS4, TS4, and FS have been presented. Source: Adapted from ref [73] with permission. Copyright 2012, published Elsevier.

optimized reduction conditions (250 °C for 2 h), a suitable proportion of active sites—namely  $\text{Cu}^0$ ,  $\text{Cu}^+$ , and  $\text{CuCr}_2\text{O}_4$ —was formed on the catalyst surface, as shown in Fig. 8. Among these,  $\text{Cu}^0$  is primarily responsible for the dehydrogenation of ethanol to acetaldehyde, while  $\text{Cu}^+$  and  $\text{CuCr}_2\text{O}_4$  provide Lewis acidic sites that promote the further reaction of acetaldehyde with ethanol to form ethyl acetate. Furthermore, the introduction of Cr significantly enhanced copper dispersion, facilitated the reduction of  $\text{Cu}^{2+}$  to  $\text{Cu}^+$  and  $\text{Cu}^0$ , and increased Lewis acidity through the formation of the  $\text{CuCr}_2\text{O}_4$  phase, thereby improving the selectivity toward ethyl acetate.

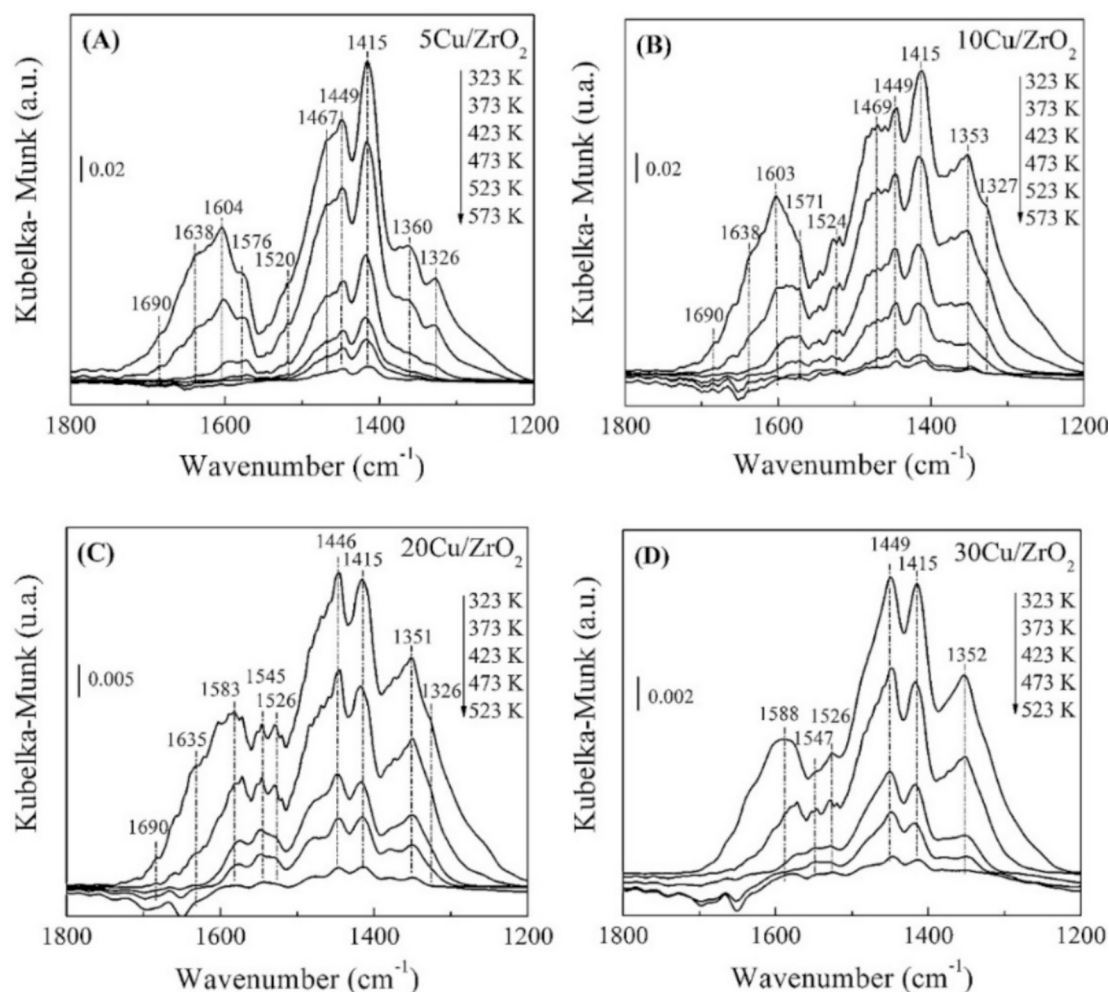
$\text{Cu}/\text{ZrO}_2$  catalysts exhibit superior performance in the selective conversion of ethanol to ethyl acetate, but their performance is closely related to the Cu loading, the dispersion state of Cu species, and the metal-support interface properties. The study by Freitas et al. [76] found that the Cu loading on  $\text{Cu}/\text{ZrO}_2$  catalysts can modulate the structure of active sites and the surface properties, thereby governing the reaction pathway selection for ethanol dehydrogenation. As the Cu loading

increased from 5 wt% to 30 wt%, the  $\text{Cu}^0$  particle size increased from 3.1 nm to 12.6 nm, and the dispersion decreased from 34 % to 8 %. When the Cu loading was moderate ( $\geq 10$  wt%), the catalyst formed active centers rich in  $\text{Cu}^0$  nanoparticles and interfacial  $\text{Cu}^+$  species. At this point, the DRIFTS of CO adsorption spectra showed a strong characteristic peak for monodentate carbonate at  $1360\text{ cm}^{-1}$  (Fig. 9), confirming the presence of a high density of strongly basic  $\text{O}^{2-}$  sites on the support surface. This multifunctional active structure acts synergistically to preferentially and efficiently convert ethanol to ethyl acetate via the acetaldehyde intermediate. In contrast, at low Cu loading (5 wt%), highly dispersed  $\text{Cu}^+$  species, together with the exposed strong Lewis acid sites ( $\text{Zr}^{4+}$ ) on the support—which correspond to the polydentate carbonate peaks observed at  $1449$  and  $1415\text{ cm}^{-1}$  in the DRIFTS spectra (Fig. 9)—promoted the formation of acetaldehyde and its subsequent side reactions via Aldol condensation.

The studies discussed above collectively demonstrate that the performance of Cu-based catalysts in the ethanol dehydrogenation to ethyl



**Fig. 8.** A proposed mechanism for the dehydrogenation of ethanol on Cu-Cr catalyst. Source: Adapted from ref [32] with permission. Copyright 2022, published Elsevier.



**Fig. 9.** FTIR spectra of CO adsorption in the LF region of 1800–1200  $\text{cm}^{-1}$  on reduced Cu/ZrO<sub>2</sub> samples: 5Cu/ZrO<sub>2</sub> (A), 10Cu/ZrO<sub>2</sub> (B), 20Cu/ZrO<sub>2</sub> (C) and 30Cu/ZrO<sub>2</sub> (D). Source: Adapted from ref [76] with permission. Copyright 2014, published Elsevier.

acetate reaction does not rely on a single type of active site, but rather stems from a synergistic system comprising multiple sites, such as Cu<sup>0</sup>, Cu<sup>+</sup>, and metal oxide supports. Although the Cu–Cr and Cu/ZrO<sub>2</sub> catalyst systems differ in their chemical composition, they both follow a similar underlying catalytic principle: Cu<sup>0</sup> sites are primarily responsible for the initial dehydrogenation step of ethanol, while the Lewis acid sites provided by Cu<sup>+</sup>, CuCr<sub>2</sub>O<sub>4</sub>, or the ZrO<sub>2</sub> support activate the key intermediate, acetaldehyde, and promote its subsequent coupling reaction with ethanol and/or facilitate the desorption of the ethyl acetate product.

Preedavijitkul [77] also investigated the deactivation of Cu–Cr catalysts over protracted ethanol dehydrogenation. They found that there are two reasons for catalyst deactivation: the first reason is because the hydrogen self-reduction process reduces CuCr<sub>2</sub>O<sub>4</sub> and Cu<sup>+</sup> active sites to Cu<sup>0</sup>. Another reason is that soft coke formed on the catalyst surface during the reaction process cover the active sites of the catalyst. This study indicates that the key to achieving long-term catalyst stability lies in stabilizing the Lewis acid sites and preventing the formation of soft deposits. Zhang et al. [78] developed an efficient CuCeZr solid solution catalyst, and the optimal CuCeZr catalyst (10 % Ce) showed 55.9 % ethanol conversion and 87.5 % ethyl acetate selectivity, and remained stable over 1000 h of continuous reaction. As shown in Fig. 10, Ce doping plays a key role in the selective conversion of ethanol to ethyl acetate. Ce doping promotes the formation of Cu<sup>+</sup> species, which acts synergistically with Cu<sup>0</sup> to promote the generation of ethyl acetate. In addition, Ce doping inhibited the formation of strong acid sites and

strong base sites, which inhibited the condensation of acetaldehyde to produce butanol and butanone.

Additionally, Xiaofei et al. [79] introduced Mn promoter into Cu–Cr–Zr composite oxide catalysts to study the effect of Mn promoter on the catalyst structure and catalytic performance. It was shown that the Mn promoter significantly improved the thermal stability and resistance to carbon deposition of the catalyst by improving the dispersion of Cu components on the catalyst surface and inhibiting the growth of CuO grains. The results of thermogravimetric experiments showed that after 140 h of reaction, the Mn-added catalyst showed almost no carbon deposition, while the surface of the non-Mn-added catalyst showed obvious carbon deposition.

Furthermore, Li et al. [80] investigated the effect of regulating the ZrO<sub>2</sub> crystal phase in CuCeZrO catalysts on their performance in the dehydrogenative condensation of ethanol to ethyl acetate. It was found that controlling the calcination temperature (500–800 °C) induced a phase transformation of ZrO<sub>2</sub>: As shown in Fig. 11, at temperatures > 600 °C, the tetragonal phase (t-ZrO<sub>2</sub>) gradually transformed into the monoclinic phase (m-ZrO<sub>2</sub>), with the proportion of m-ZrO<sub>2</sub> increasing from 0 % at 500 °C to 95 % at 800 °C. As the m-ZrO<sub>2</sub> content increased, the density of strong basic sites in the catalyst decreased progressively, significantly suppressing the condensation of acetaldehyde. Consequently, the selectivity for ethyl acetate improved from 68 % (CCZ-500) to 78 % (CCZ-700).

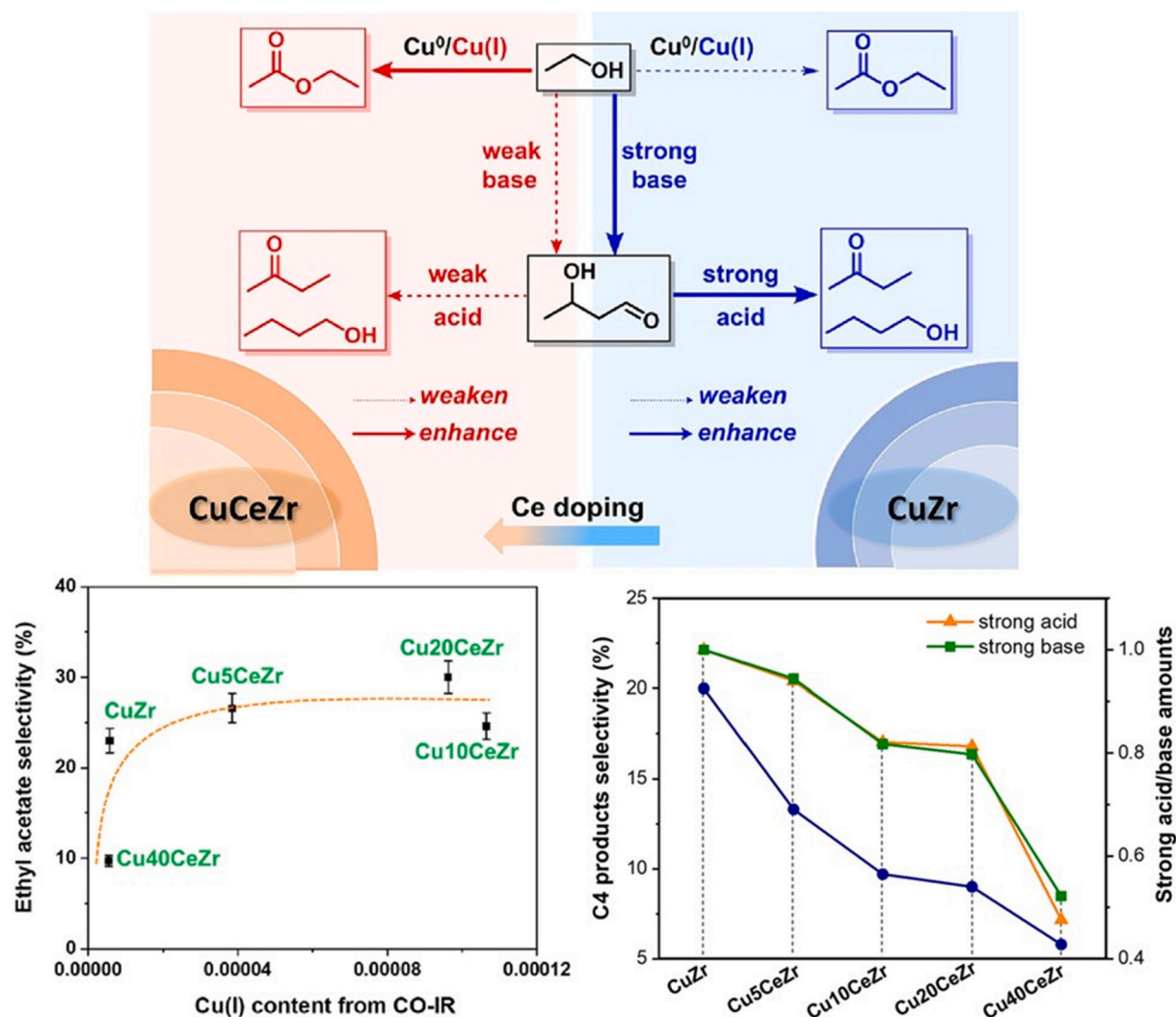


Fig. 10. Proposed reaction pathways before and after Ce doping into CuZr catalyst ; Relationship between ethyl acetate selectivity and Cu (I) content from CO-IR ; The relationship between C4 by-products selectivity and strong acid-base sites. Source: Adapted from ref [78] with permission. Copyright 2023, published Elsevier.

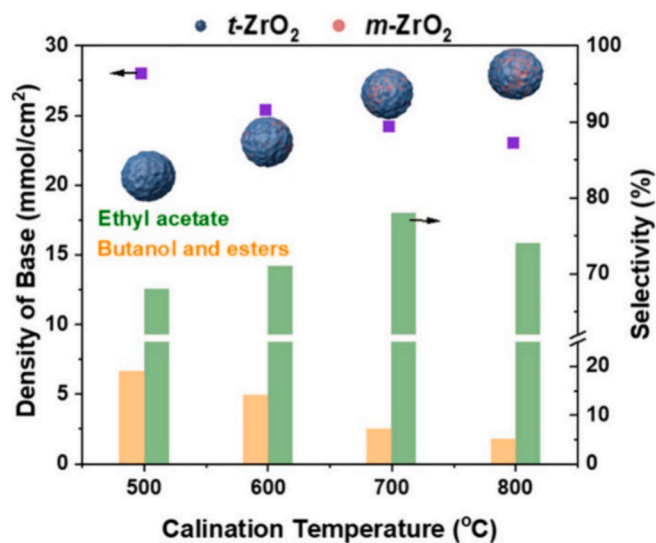


Fig. 11. Relationship of the crystal phase, density of the strong base, ethyl acetate selectivity, butanol and esters selectivity, and calcination temperature over CuCeZrO catalysts. Source: Adapted from ref [80] with permission. Copyright 2014, published American Chemical Society.

### 3.2. Pd-based catalysts

Studies have shown that supported Pd catalysts achieve high activity and selectivity through unique electronic structure modulation and intermediate stabilization mechanisms [81–84].

On monometallic Pd catalysts, Pd<sup>0</sup> not only acts as the active site for ethanol dehydrogenation but also facilitates the further decomposition of acetaldehyde into C1 products (such as CH<sub>4</sub> and CO), thereby reducing the selectivity for the target product. Therefore, inhibiting ethanol decomposition is crucial for the selective conversion of ethanol to ethyl acetate. Iwasa et al. [81] demonstrated that palladium-based catalysts, by forming alloys with metals such as zinc, gallium, and indium (Pd–Zn, Pd–Ga, and Pd–In), can inhibit the ethanol decomposition reaction by stabilizing the acetaldehyde intermediate and modifying the electronic structure of the Pd surface, thereby promoting the further conversion of the intermediate into ethyl acetate.

Sánchez et al. [83] designed a Pd–Zn catalytic membrane reactor (Pd–Zn–CMR) that synergistically combines a selective Pd–Zn alloy catalyst with a H<sub>2</sub>-permeable membrane (CeO<sub>2</sub>–ZrO<sub>2</sub>/TiO<sub>2</sub>) to enhance ethyl acetate production. The membrane operates in a flow-through mode under Knudsen diffusion, which preferentially removes the co-product H<sub>2</sub> from the reaction zone. This shifts the reaction equilibrium ( $2\text{CH}_3\text{CH}_2\text{OH} \rightleftharpoons \text{CH}_3\text{COOCH}_2\text{CH}_3 + 2\text{H}_2$ ), thereby overcoming thermodynamic limitations and boosting ethanol conversion. Concurrently,

the PdZn alloy component increases selectivity by diluting Pd<sup>0</sup> sites, suppressing undesired C–C bond cleavage. This system exhibited excellent stability over 5 h at 250 °C and 1 MPa.

### 3.3. Other types of catalysts

Despite the excellent catalytic activity and stability of CuCr-based catalysts, the toxicity of Cr and the low stability of Cu particles limit their practical applications. Moreover, Cu-based catalysts frequently undergo side reactions, such as C–C bond cleavage and condensation that lead to product complexity and catalyst deactivation.

For this purpose, Yin et al. [85] chemically tailored Nickel-based catalyst (Ni-based) catalysts by introducing In, and successfully developed the Ni<sub>0.1</sub>Zn<sub>0.7</sub>Al<sub>0.3</sub>InO<sub>x</sub> catalyst for the dehydrogenative coupling of ethanol into ethyl acetate. Under conditions of 275 °C and a weight hourly space velocity (WHSV) of 1.9 h<sup>−1</sup>, the catalyst achieved 46.2 % ethanol conversion and 90.1 % ethyl acetate selectivity, while maintaining stable performance during a 370-hour time on stream. It was shown that the introduction of In led to the formation of Ni<sub>4</sub>In alloy active sites, which effectively inhibited the C–C bond cleavage and condensation side reactions during ethanol conversion by modulating the electronic properties of Ni. DFT calculations further confirmed that the energy barrier for acetaldehyde dehydrogenation to acetyl species on the Ni<sub>4</sub>In(110) crystal surface was 0.41 eV, significantly lower than that of the 1.89 eV on the pure Ni(111) surface, elucidating the efficient selective regulation mechanism of alloy sites in dehydrogenation reactions.

In addition, Mück et al. [86] prepared transition metal (Co, Ni, Cu, Mn, Cr) promoted Mg-Fe mixed oxide catalysts by the coprecipitation method for the gas-phase conversion of ethanol to ethyl acetate and other products. The introduction of Cu and Ni significantly enhanced the dehydrogenation activity and promoted ethanol conversion to ethyl acetate via the Tishchenko reaction pathway. Among these, the Cu–Mg–Fe catalysts exhibited optimal performance at 300–350 °C, achieving ethanol conversions of 65–70 %. However, Ni promoted C–C bond cleavage at high temperatures (350 °C), leading to increased gas-phase byproducts such as CH<sub>4</sub> and CO<sub>2</sub>. The Co–Mg–Fe catalysts are moderately active but prone to the formation of gaseous by-products (e. g., ethylene, propane, and butane) at high temperatures (350 °C). In contrast, Mn- and Cr-modified catalysts performed poorly: despite the higher density of acid-base sites on the surface of Mn, its dehydrogenation capacity was limited, while Cr exhibited extremely low ethanol conversion (<1%) across the entire temperature range (280–350 °C) due to insufficient redox activity.

## 4. Dehydrogenative coupling of ethanol to n-butanol

N-butanol, for its part, has been recognized as a sustainable fuel with advantages over ethanol due to its higher energy density and excellent fuel properties [87–91]. The conventional industrial production of n-butanol relies on fossil-based feedstocks (e.g., carbonylation of fossil-based propylene), yet it faces challenges related to resource depletion and environmental burdens. The Guerbet reaction provides a green alternative route, converting renewable ethanol into n-butanol while simultaneously addressing the issue of bioethanol overcapacity (laboratory stage) [34]. As shown in Scheme 1, this process consists of four key consecutive reaction steps: (1) Firstly, ethanol undergoes dehydrogenation to form acetaldehyde; (2) secondly, acetaldehyde undergoes aldol condensation to form 3-hydroxybutanal; (3) thirdly, 3-hydroxybutanal undergoes dehydration to form crotonaldehyde; and (4) finally, crotonaldehyde undergoes hydrogenation to form n-butanol. This review systematically analyzes the catalytic performance of various catalysts for the dehydrogenative coupling of ethanol to n-butanol, as summarized in Table 4.

**Table 4**

Catalysts for dehydrogenative coupling of ethanol to n-butanol.<sup>a</sup>

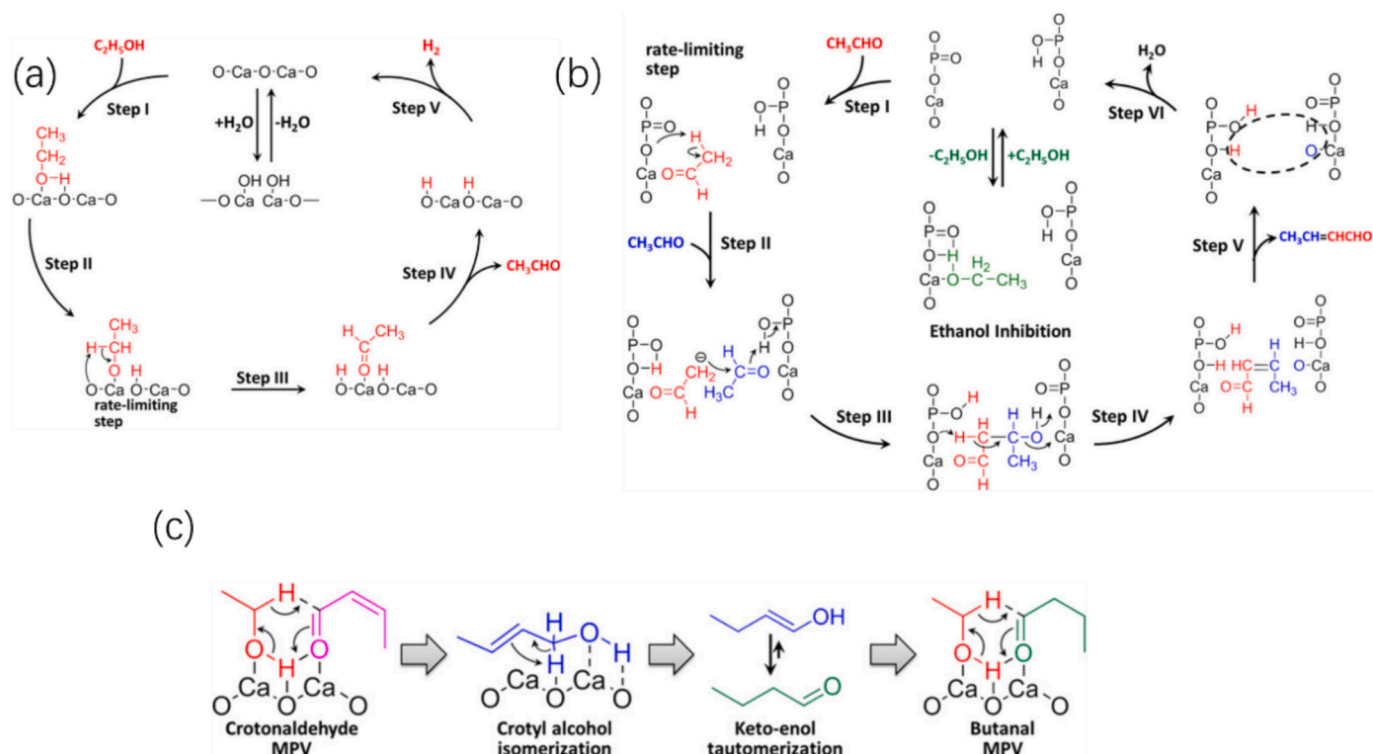
| Catalyst                                                            | Reaction conditions                                                                     | Conv.                            | Sel.                             | Feature                                                                                                                          | Ref. |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------|
| Cu–CeO <sub>2</sub><br>Pd<br>–CeO <sub>2</sub>                      | N <sub>2</sub> , 250 °C, 2 MPa, LHSV = 4 mL/(h·g-cat), N <sub>2</sub> /ethanol = 500:1, | Cu: 46.2 %<br>Pd: 11.9 %<br>34 % | Cu: 41.2 %<br>Pd: 67.6 %<br>68 % | Metal-support synergism, Cu promotes ethanol dehydrogenation, Pd enhances catalytic activity and selectivity                     | [91] |
| Pt-Y/Beta                                                           | N <sub>2</sub> , WHS = 2.0 h <sup>−1</sup> , 200 °C, 0.1 MPa                            |                                  |                                  | Pt is responsible for dehydrogenation and hydrogenation, Y promotes acetaldehyde condensation.                                   | [34] |
| Ni/La <sub>2</sub> O <sub>3</sub> /γ-Al <sub>2</sub> O <sub>3</sub> | N <sub>2</sub> , WHSV = 0.79, 250 °C, 100 bar                                           | 41 %                             | 51 %                             | Bimetallic Ni-Co synergistically enhances conversion, with ethanol dehydrogenation as the rate-determining step.                 | [97] |
| Pd@UiO-66                                                           | N <sub>2</sub> , 250 °C, 2 MPa, LHSV = 4 mL/(h·g-cat), N <sub>2</sub> /ethanol = 250:1  | 49.8 %                           | 48.6 %                           | Pd nanoparticles and UiO-66 coordinated Zr sites synergistically catalyze, enhancing dehydrogenation and condensation abilities. | [89] |
| Cu@UiO-66                                                           | N <sub>2</sub> , 280 °C, 2 MPa, LHSV = 4 mL/(h·g-cat))                                  | 26.8 %                           | 60.3 %                           | Zr nodes contain Lewis acid-oxygen vacancy pairs (Zr <sub>3</sub> □)                                                             | [94] |
| Ni-MgAlO                                                            | N <sub>2</sub> , 250 °C, 3 MPa, WHSV = 3.2 h <sup>−1</sup>                              | 18.7 %                           | 55.2 %                           | Ni promotes ethanol dehydrogenation, and acid-base sites facilitate acetaldehyde condensation.                                   | [96] |
| Ag/MgAl-LDO                                                         | N <sub>2</sub> , 350 °C, 0.1 MPa, LHSV = 6 mL/(h·g-cat)                                 | 23.2 %                           | 76.6 %                           | Ag-layered oxide provides dehydrogenation sites, and acid-base synergism promotes condensation reactions.                        | [90] |
| Cu–CeO <sub>2</sub> /AC                                             | batch reactor, 250 °C, 0.1 MPa N <sub>2</sub> , reaction for 48 h                       | 39.1 %                           | 55.2 %                           | Synergistic effect of Cu, CeO <sub>2</sub> , and AC                                                                              | [95] |

<sup>a</sup> Reactions were performed in a fixed bed reactor if not specified; Conv. and Sel. are abbreviations for ethanol conversion and n-butanol selectivity, respectively.

### 4.1. Active sites and structure-activity relationships in the guerbet reaction

The reaction pathways and mechanisms of the Guerbet reaction varies significantly depending on the catalyst system employed. Regarding the active sites governing the condensation step, some studies emphasize the primary role of basic sites, considering acid sites to function mainly as auxiliary stabilizers. However, other research has revealed that Lewis acid sites can also play a dominant role in this step.

The study by Ho et al. [33] elucidated the mechanism of ethanol coupling to butanol over a hydroxyapatite (HAP) catalyst via the Guerbet pathway. This process is synergistically catalyzed by two distinct types of active sites (Scheme 4): Ca–O basic sites are responsible for the dehydrogenation of ethanol to acetaldehyde and subsequent hydrogen transfer steps, while CaO/PO<sub>4</sub><sup>3−</sup> acid–base pairs collectively catalyze the aldol condensation of acetaldehyde. During the condensation process, the basic oxygen in PO<sub>4</sub><sup>3−</sup> abstracts the α-hydrogen of acetaldehyde to form a key enolate intermediate, which is the rate-determining step of the overall reaction. In addition, weak Lewis acid



**Scheme 4.** (a) Reaction Mechanism for Ethanol Dehydrogenation to Acetaldehyde over HAP. (b) Reaction Mechanism for Aldol Condensation of Acetaldehyde to Crotonaldehyde over HAP. (c) Hydrogen Transfer Pathway between Crotonaldehyde and Ethanol. Source: Adapted from ref [33] with permission. Copyright 2016, published American Chemical Society.

sites (e.g.,  $\text{Ca}^{2+}$  and  $\text{P}-\text{OH}$ ) facilitate C–C bond formation by stabilizing adsorbed acetaldehyde molecules and the negatively charged enolate intermediate, demonstrating the synergistic catalytic role of acid and base sites in the reaction. Furthermore, Hanspal et al. [35] elucidated the microscopic mechanism behind the performance differences between hydroxyapatite (HAP) and magnesia (MgO) in the Guerbet coupling of ethanol to butanol using multiproduct steady-state isotopic transient kinetic analysis (SSITKA). The SSITKA results revealed that the surface residence times of reactive intermediates leading to butanol ( $\tau_{\text{enOH}}$ ) were significantly longer than those leading to acetaldehyde ( $\tau_{\text{aCH}}$ ) over both catalysts, confirming that C–C coupling is the rate-determining step. However, the surface properties of the catalysts dictate the reaction pathway of the key acetaldehyde intermediate: HAP, with its high density of weak base–weak Lewis acid site pairs, effectively “locks” the acetaldehyde intermediate on the surface, favoring its conversion to butanol via a surface reaction path. In contrast, the lack of acid sites and excessively strong basicity on MgO lead to extensive desorption of acetaldehyde, thereby limiting C–C coupling.

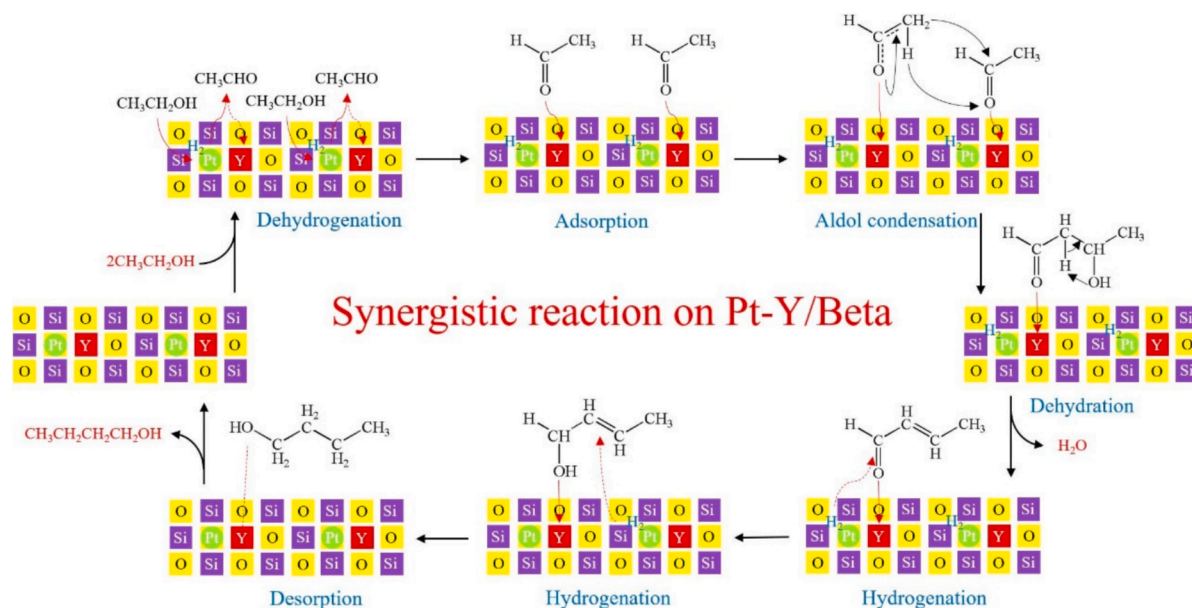
Liu et al. [34] developed a bifunctional Pt-Y/Beta catalyst by incorporating yttrium (Y) into the framework of Beta zeolite in a tetra-coordinated fashion, forming Y–O–Si structured Lewis acid sites, while supporting platinum (Pt) as highly dispersed nanoparticles on the zeolite surface. The key to the catalyst’s performance lies in the precise control of acid site type to steer the reaction pathway: The original Si–OH–Al bridging hydroxyl groups in H-Beta zeolite act as Brønsted acid sites, which promote ethanol dehydration to ethylene—an undesired competing side reaction. In contrast, the introduced Y species generate Lewis acid sites due to the electronegativity difference between Y and Si, creating electron-deficient centers on Y atoms. These Lewis acid sites serve as the active centers for the aldol condensation step, activating acetaldehyde molecules to facilitate  $\alpha$ -H deprotonation and subsequent C–C coupling with another acetaldehyde molecule, yielding 3-hydroxybutanal. This step constitutes the crucial C–C bond formation in the Guerbet reaction pathway. Meanwhile, the Pt sites are responsible for

both ethanol dehydrogenation and hydrogenation of intermediates (such as crotonaldehyde), working in synergy with the Y sites to drive the cascade reaction efficiently. Based on experimental analysis, they proposed a reaction mechanism for the synthesis of n-butanol from ethanol on Pt-Y/ $\beta$  catalysts, as shown in Scheme 5.

Ordonsky et al. [92] investigated the reaction mechanism of acetaldehyde condensation to crotonaldehyde over a  $\text{ZrO}_2/\text{SiO}_2$  catalyst. By contrast to the study by Liu et al. [34], their work identified the involvement of Brønsted acid sites (Si–OH) in activating the carbonyl group of acetaldehyde. The proposed mechanism is as follows: the carbonyl group of acetaldehyde molecule A coordinates with a Lewis acid site ( $\text{Zr}^{4+}$ ), which not only polarizes the carbonyl group but also facilitates the deprotonation of its  $\alpha$ -hydrogen. Concurrently, acetaldehyde molecule B is activated by forming a strong hydrogen bond with a Brønsted acid site (Si–OH). Subsequently, with the assistance of an adjacent weak base ( $\text{Zr}-\text{O}-$ ), the  $\alpha$ -hydrogen of molecule A is abstracted, generating a nucleophile that attacks the carbonyl carbon of the activated molecule B, leading to the formation of a 3-hydroxybutanal intermediate via nucleophilic addition.

Additionally, Jiang et al. [89] found that in the Pd@UiO-66 catalyst for the ethanol to n-butanol reaction, the Lewis acid sites (Zr–CUSs), rather than the traditional Lewis base sites, play a dominant role in the critical aldol condensation step. Kinetic analysis revealed that the apparent activation energy for aldol condensation over this catalyst (55.8 kJ/mol) was lower than that of traditional catalysts like Cu–CeO<sub>2</sub>/AC (70.1 kJ/mol), indicating the high efficiency of the Zr–CUSs. Furthermore, selective poisoning experiments demonstrated that blocking the acid sites (Zr–CUSs) significantly suppressed the activity, whereas poisoning the base sites had a minimal effect.

The aforementioned studies on the Guerbet reaction collectively reveal that its reaction pathway is highly dependent on the surface properties of the catalyst, with the core challenge lying in efficiently catalyzing the rate-determining aldol condensation step. Although the dominant active sites vary across different catalytic systems—such as



**Scheme 5.** Synergistic reaction mechanism for the synthesis of n-butanol from ethanol on Pt-Y/Beta. Source: Adapted from ref [34] with permission. Copyright 2024, published Elsevier.

HAP, Pt-Y/Beta, and  $\text{ZrO}_2/\text{SiO}_2$ —ranging from basic sites to Lewis acids or even Brønsted acids, the essence of their efficient catalysis stems from the sophisticated synergy between acid and base sites. This synergistic interaction effectively stabilizes reaction intermediates and reduces the energy barrier of key steps, thereby governing the reaction pathway and ultimate product selectivity. Consequently, future catalyst development is better to focus more on the precise construction and regulation of bifunctional active centers with synergistic effects, which is crucial for achieving high efficiency in the Guerbet reaction and other multi-step cascade reactions.

#### 4.2. Noble metal catalysts

The superior performance of supported noble metal catalysts for ethanol-to-butanol conversion benefits from synergistic catalysis among active sites and support properties like the isolation effect and tunable acidity/basicity.

Wu et al. [91] prepared M-CeO<sub>2</sub>/activated carbon (AC) (M = Cu, Fe, Co, Ni, Pd) catalysts via the wetness impregnation method and evaluated their performance in the catalytic upgrading of ethanol to n-butanol under mild conditions (250 °C, 2 MPa, N<sub>2</sub>/ethanol (v/v) = 500:1). The results showed that the Pd-CeO<sub>2</sub>/AC catalyst exhibited the highest n-butanol selectivity (67.6 %), while the Cu-CeO<sub>2</sub>/AC catalyst achieved the highest ethanol conversion (46.2 %) with moderate n-butanol selectivity (41.3 %). This was attributed to the strong hydrogen activation and spillover ability of Pd metal, which significantly promoted the CeO<sub>2</sub> reduction and the formation of surface basic sites to enhance the crotonaldehyde hydrogenation efficiency. In contrast, Cu metal dominated ethanol conversion due to its superior dehydrogenation capability.

Furthermore, the Pd@UiO-66 catalyst [89] (2 wt% Pd) discussed in the previous section exhibits excellent performance in the conversion of ethanol to n-butanol. Under conditions of 250 °C and 2 MPa, it achieves an ethanol conversion rate of 49.8 % and an n-butanol selectivity of 48.6 %, while maintaining stability over a long reaction time of 200 h. This is attributed to the high activity of the palladium active sites in the steps of ethanol dehydrogenation and crotonaldehyde hydrogenation, as well as the Lewis acidic Zr<sub>6</sub> nodes, which promote the aldol condensation of acetaldehyde.

Zhang et al. [90,93] investigated the properties of Ag particles

supported on MgAl-layered double oxides (Ag/MgAl-LDO) in the dehydrogenation coupling of ethanol to n-butanol. It was shown that synergistic catalysis between the acid-base bifunctional sites on the LDO surface and the Ag particles: the basic sites promote the adsorption and subsequent dehydrogenation of ethanol on Ag particles to form acetaldehyde, while the acid-base cooperation further promotes the aldol condensation of acetaldehyde to form C-C bonds. The Ag/MgAl-LDO catalyst (Mg/Al = 2) achieved 23.2 % ethanol conversion and 76.6 % n-butanol selectivity at 350 °C and 0.1 MPa. At 250 °C and 2 MPa, the ethanol conversion reached 45.2 % while maintaining a n-butanol selectivity of 60.3 %.

Recently, Liu et al. [34] developed a bifunctional Pt-Y/Beta catalyst by embedding yttrium (Y) and platinum (Pt) species into the zeolite framework and surface by solid-state ion exchange and impregnation using H-Beta zeolite as the support. The catalyst achieved 33 % ethanol conversion and 68 % n-butanol selectivity under reaction conditions at 200 °C. The active sites and reaction mechanism for the conversion of ethanol to n-butanol on Pt-Y/Beta catalysts are discussed in Section 4.1.

#### 4.3. Non-noble metal catalysts

Traditional noble metal catalysts exhibit excellent activity in the ethanol dehydrogenative coupling to n-butanol, but their high cost and resource scarcity limit industrial applications. In contrast, non-noble metal catalysts have attracted attention due to their low cost, abundant resources, and structural tunability.

Zhou et al. [94] precisely engineered Cu@UiO-66 catalysts, successfully revealing the critical role of spatial proximity between active sites in the tandem catalytic upgrading of ethanol to n-butanol. The study identified that Cu nanoparticles confined within the pores of UiO-66 and the Lewis acid-oxygen vacancy pairs (Zr<sub>3</sub>-□) on Zr nodes serve as the genuine active centers. The close contact between them efficiently synergizes to catalyze three key steps: ethanol dehydrogenation, acetaldehyde condensation, and crotonaldehyde hydrogenation. Benefiting from this unique structure, the catalyst achieved a high n-butanol yield of 22.2 % at 280 °C and 2 MPa, while significantly suppressing side reactions (C-C bond cleavage and ethanol dehydration) and delivering an exceptionally high liquid product yield of 98.1 %.

Jiang et al. [95] developed a Cu-CeO<sub>2</sub>/activated carbon (AC) catalyst for the efficient and continuous catalytic upgrading of ethanol to n-

butanol under mild conditions (250 °C, 2 MPa). Its outstanding performance originates from the clever synergistic effects among Cu, CeO<sub>2</sub>, and the AC support: the highly dispersed Cu nanoparticles are responsible for the dehydrogenation of ethanol and the re-hydrogenation of intermediates; the highly dispersed CeO<sub>2</sub> species not only provide the basic sites required for the crucial aldol condensation step but also effectively inhibit the sintering of Cu particles through strong interaction with Cu, thereby significantly enhancing the catalyst's stability; meanwhile, the AC support greatly promotes the high dispersion of the active components (Cu and CeO<sub>2</sub>) due to its high specific surface area. This synergy enables the catalyst to achieve a high n-butanol yield of up to approximately 20 % while avoiding C–C bond cleavage and the formation of by-products.

Pang et al. [96] successfully prepared a highly dispersed Ni–MgAlO catalyst with 19.5 wt% Ni loading through a layered double hydroxide (LDH) precursor-derived strategy, achieving continuous ethanol upgrading to value-added n-butanol and C<sub>4</sub>–C<sub>8</sub> alcohols in a fixed-bed reactor. It was shown that Ni metal sites dominate the dehydrogenation of ethanol to produce acetaldehyde and reactive hydrogen species, whereas strong basic sites provided by MgO and moderate acidic sites derived from Al<sub>2</sub>O<sub>3</sub> promote aldol condensation (to produce β-hydroxybutyraldehyde) and dehydration (to form crotonaldehyde), respectively. The optimal Ni<sub>4</sub>MgAlO catalyst exhibited high selectivity (n-butanol selectivity reached 55.2 %) and excellent operational stability (no activity decay over 60 h) under mild conditions at 250 °C, 3 MPa hydrogen pressure, and a weight hourly space velocity (WHSV) of 4 h<sup>−1</sup>.

In summary, although these three studies employ different catalytic systems, they share the common key strategy of constructing spatially proximate and functionally cooperative multi-active sites to efficiently drive the tandem reaction of ethanol to n-butanol. These studies have made valuable contributions to the high-value conversion of ethanol using non-noble metal catalysis.

## 5. Conclusion and outlook

Bioethanol dehydrogenation and dehydrogenation coupling to produce high-value-added products are important directions in current green chemistry and catalysis research, and products such as acetaldehyde, ethyl acetate, n-butanol, and by-product hydrogen show a wide range of application potential and value in the chemical industry. This review focuses on three main products acetaldehyde, ethyl acetate, and n-butanol, and systematically reviews the research progress in ethanol dehydrogenation and dehydrogenative coupling reactions. The details include the relationship between catalyst structure and performance, selectivity manipulation, and mechanism modulation. As shown in the present review, recent years have seen significant breakthroughs in catalyst design and synthesis, optimization of reaction conditions, and catalytic performance enhancement for the three products. However, challenges and technical bottlenecks remain regarding catalyst activity and selectivity, catalyst stability, process conditions and energy consumption, and environmental and economic concerns, requiring further research and resolution.

### 5.1. Catalyst activity and selectivity issues

Balancing activity and selectivity has long been a fundamental challenge in catalyst design. The selectivity toward acetaldehyde, ethyl acetate, and n-butanol is co-regulated by multiple factors, including the nature of active sites, the acidity/basicity of the support, and hydrogen concentration [20,21,74,98–100]. On Cu-based catalysts, for example, the Cu<sup>+</sup> and Cu<sup>0</sup> sites play a synergistic role in the generation of different products. Cu<sup>+</sup> can activate O–H in the ethanol molecule, while Cu<sup>0</sup> is responsible for activating the C–H bond, thus enabling the dehydrogenation of ethanol to produce acetaldehyde [28,29,101,102]. During the formation of ethyl acetate, Cu<sup>0</sup> promotes the dehydrogenation of ethanol to acetaldehyde, and subsequently Cu<sup>+</sup> promotes the coupling

reaction of acetaldehyde [32,73]. The formation of n-butanol depends on Cu<sup>0</sup> to supply acetaldehyde, while Cu<sup>+</sup> acts as a Lewis acid site to facilitate the aldol condensation of acetaldehyde, thereby forming intermediates for C–C coupling [103]. The details of how the electronic structures of Cu<sup>+</sup> and Cu<sup>0</sup> influence the activation of O–H and C–H bonds will be discussed in the main text. Future studies should employ more in situ DRIFT and DFT calculations to investigate the reaction mechanisms, identify key active sites and critical intermediates, and determine the rate-determining steps. This will aid in the rational regulation of the electronic properties of active sites and the metal–support interface, thereby guiding the design of highly efficient catalysts [103–106].

### 5.2. Catalyst stability strategies

The long-term stability of catalysts is one of the major challenges in their practical applications [63,77,107–111]. In ethanol dehydrogenation and dehydrogenation coupling reactions, some catalysts exhibit poor stability under reaction conditions, with some even struggling to operate continuously for more than 100 h. For copper-based catalysts, due to their low Tammann temperature (approximately 405 °C), copper is prone to migration and agglomeration during the reaction process, ultimately leading to sintering and deactivation [24,53]. Strategies such as structural confinement and chemical anchoring can be employed to address these issues. For instance, nanoscale spatial structures (such as silicalite-1 and mesoporous silica spheres) can be used to physically confine and isolate active sites [14,27]. Alternatively, inexpensive ligands like ethylenediamine and propanediamine can form complexes with copper, thereby uniformly “anchoring” the active metal onto the support surface [50]. Additionally, introducing metal promoters to leverage synergistic effects with the active components is also an effective approach to stabilizing catalysts. For example, the introduction of Ni single atoms in NiCu alloys significantly enhances the stability of Cu [58]. Similarly, adding Mn as a promoter in Cu-based catalysts helps improve the dispersion of Cu species and suppresses the growth of CuO crystallites [79].

### 5.3. Process conditions and energy consumption

Currently, thermocatalytic ethanol dehydrogenation and dehydrogenative coupling reactions under laboratory conditions typically require high temperatures and pressures, leading to high energy consumption and stringent equipment requirements. For example, the production of acetaldehyde using Cu/ZnO@SiO<sub>2</sub> catalyst must be carried out at 310 °C, while the synthesis of n-butanol with Pd@UiO-66 catalyst requires conditions of 250 °C and 2 MPa [24,32,64,96]. Notably, the NiCu single-atom catalyst reduces the acetaldehyde desorption temperature by 30 K and exhibits ethanol dehydrogenation activity 12-fold higher than that of pure copper. Meanwhile, the PtCu single-atom catalyst activates the O–H and C–H bonds of ethanol at 160 K and 310 K, respectively [58,59]. By precisely modulating the electronic state of the catalyst surface to minimize reaction energy barriers, it may be possible to address this issue effectively. In addition, conventional fixed-bed reactors and reaction vessels face inherent limitations, primarily thermodynamic equilibrium constraints and the high energy consumption associated with downstream product separation. The use of membrane reactors for in situ H<sub>2</sub> removal can shift the reaction equilibrium, enabling higher conversions under milder conditions compared to thermal processes [83]. Furthermore, the development of highly efficient photocatalytic or electrocatalytic systems represents a promising pathway. However, critical challenges remain in the creation of core materials that combine high activity, stability, and low cost, which must be urgently addressed [112–115]. Finally, enhancing the comprehensive utilization of by-product hydrogen—for instance, by coupling it with downstream hydrogenation reactions or directly applying it in energy conversion systems—can

improve overall energy efficiency and advance green and sustainable development [101].

#### 5.4. Environmental and economic issues

Although bioethanol is a green and renewable resource, the preparation of catalysts often involves toxic promoters (e.g.,  $\text{Cr}_2\text{O}_3$ ,  $\text{Cr}(\text{NO}_3)_3$ ) and corrosive reagents (such as metal nitrates,  $\text{H}_2\text{SO}_4$ , and  $\text{NaOH}$ ), leading to the generation of harmful waste. Moreover, current laboratory processes generally operate under high-temperature and high-pressure conditions. Scaling up such processes would require careful evaluation of both carbon emissions and economic feasibility [63,116]. Future research should prioritize exploring more environmentally friendly approaches, including gaining deeper insights into reactive species, structure–activity relationships, and reaction mechanisms, as well as developing greener and more cost-effective reagents and catalyst synthesis methods to establish stable and efficient catalytic systems. To replace toxic Cr-based promoters, In-doped Ni based catalysts or Cu and Ni doped Mg–Fe mixed oxide catalysts can be used for the conversion of ethanol to ethyl acetate [85,86]. Additionally, employing silica, biochar, or similar materials as supports in the dehydrogenation of ethanol to acetaldehyde not only suppresses side reactions (e.g., dehydration and condensation) but also helps reduce costs [27,49]. By utilizing strategies such as doping and alloying, the performance of non-precious metal catalysts (e.g., Cu, Ni) can be enhanced, thereby decreasing the dependence on scarce precious metals such as Au, Ag, and Pt [25,26].

#### CRedit authorship contribution statement

**Qingwei Meng:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Fucheng Chen:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Xin Li:** Writing – original draft, Conceptualization. **Chengwu Qiu:** Writing – review & editing. **Qingchun Yuan:** Writing – review & editing, Writing – original draft. **Tiejun Wang:** Writing – review & editing.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Acknowledgements

This work was supported by the National Key Research & Development Program of China (No. 2023YFB4203604).

#### Data availability

Data will be made available on request.

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